Introduction to Harmonic Analysis

Lecture notes, expanded

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Abstract

These notes follow the handwritten notebook page by page; the grey rules "notebook page N" mark where page N of the notebook begins. Everything said in the lectures is here, but in addition every notion that was used without definition is defined (in grey *Background* boxes), every "check this" and every exercise is worked out, and remarks are explained.

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1 Fourier analysis in Hilbert spaces

1.1 Inner products and norms

Background 1. (Vector spaces over $\mathbb{C}$)

A *linear space* (vector space) over $\mathbb{C}$ is a set H with an addition $H \times H \to H$ and a scalar multiplication $\mathbb{C} \times H \to H$ such that $(H, +)$ is an abelian group (with neutral element 0) and for all $\alpha, \beta \in \mathbb{C}$, $g, h \in H$:

$$\alpha(g+h) = \alpha g + \alpha h, \quad (\alpha+\beta)g = \alpha g + \beta g, \quad (\alpha\beta)g = \alpha(\beta g), \quad 1 \cdot g = g.$$

In particular $g, h \in H$, $\alpha, \beta \in \mathbb{C} \Rightarrow \alpha g + \beta h \in H$. For $z = a + ib \in \mathbb{C}$ we write $\overline{z} = a - ib$, $\operatorname{Re} z = a = \frac{z+\overline{z}}{2}$, $\operatorname{Im} z = b = \frac{z-\overline{z}}{2i}$ and $|z|^2 = z\overline{z}$.

Definition 1.2 (Inner product)**.** Let H be a linear space over $\mathbb{C}$. An *inner product* on H is a map $(\cdot,\cdot) : H \times H \to \mathbb{C}$ such that

(i) $(x,y) = \overline{(y,x)}$ for all $x, y \in H$ (Hermitian symmetry)

(ii) $(x,x) \geq 0$ for all $x \in H$, and $(x,x) = 0 \iff x = 0$ (positive definiteness)

(iii) $(\alpha x + \beta y, z) = \alpha\,(x,z) + \beta\,(y,z)$ for all $\alpha, \beta \in \mathbb{C}$, $x, y, z \in H$ (linearity in the first slot)

Note that (i) implies $(x,x) = \overline{(x,x)} \in \mathbb{R}$, so the inequality in (ii) makes sense.

Remark 1.3 (Conjugate linearity in the second slot). $(x, \alpha y + \beta z) = \overline{\alpha}\,(x,y) + \overline{\beta}\,(x,z)$. Indeed, by (i), (iii), (i): $(x, \alpha y + \beta z) = \overline{(\alpha y + \beta z, x)} = \overline{\alpha\,(y,x) + \beta\,(z,x)} = \overline{\alpha}\,\overline{(y,x)} + \overline{\beta}\,\overline{(z,x)} = \overline{\alpha}\,(x,y) + \overline{\beta}\,(x,z)$. (Physicists use the opposite convention: linear in the second slot.)

Definition 1.4. For $x \in H$ put $\|x\| := \sqrt{(x,x)}$.

Background 5. (Norm)

A *norm* on a linear space H is a map $\|\cdot\| : H \to [0,\infty)$ with

(N1) $\|x\| = 0 \iff x = 0$;

(N2) $\|\lambda x\| = |\lambda|\,\|x\|$ for $\lambda \in \mathbb{C}$;

(N3) $\|x + y\| \leq \|x\| + \|y\|$ (triangle inequality).

A norm induces a metric $d(x,y) = \|x - y\|$, hence notions of convergence ($x_n \to x$ iff $\|x_n - x\| \to 0$), open and closed sets, closure, etc.

Proposition 1.6. *$\|\cdot\|$ is a norm on H.*

The key step is the following inequality, which we will use constantly.

Theorem 1.7 (Cauchy–Schwarz inequality)**.** *For all $x, y \in H$,*

$$|(x,y)| \leq \|x\|\,\|y\|\,. \tag{1}$$

Equality holds if and only if x and y are linearly dependent.

Proof. If $(x,y) = 0$ there is nothing to prove; so let $(x,y) \neq 0$, in particular $x \neq 0 \neq y$. For every $\alpha \in \mathbb{C}$,

$$\begin{aligned} 0 \leq \|x + \alpha y\|^2 &= (x + \alpha y, x + \alpha y) = (x,x) + (x, \alpha y) + (\alpha y, x) + |\alpha|^2\,(y,y) \\ &= \|x\|^2 + |\alpha|^2\,\|y\|^2 + 2\,\mathrm{Re}\,(x, \alpha y)\,, \end{aligned}$$

because $(\alpha y, x) = \overline{(x, \alpha y)}$ and $w + \overline{w} = 2\,\mathrm{Re}\,w$. Choose α with $|\alpha| = 1$ such that $(x, \alpha y) = \overline{\alpha}\,(x,y) = -\,|(x,y)|$, i.e.

$$\overline{\alpha} = -\frac{|(x,y)|}{(x,y)}, \qquad \alpha = -\frac{|(x,y)|}{\overline{(x,y)}} = -\frac{|(x,y)|}{(y,x)}.$$

Then $0 \leq \|x\|^2 + \|y\|^2 - 2\,|(x,y)|$, that is

$$2\,|(x,y)| \leq \|x\|^2 + \|y\|^2 \qquad \forall x, y \in H. \tag{2}$$

This is weaker than (1) but it is not homogeneous, and we can exploit that: apply (2) to x and ty, $t > 0$:

$$2t\,|(x,y)| \leq \|x\|^2 + t^2\,\|y\|^2\,.$$

Choosing $t = \|x\| / \|y\|$ (so that $t^2 \|y\|^2 = \|x\|^2$) gives $2t\,|(x,y)| \le 2\|x\|^2$, i.e. $|(x,y)| \le \frac{1}{t}\|x\|^2 = \frac{\|y\|}{\|x\|}\|x\|^2 = \|x\|\,\|y\|$.

Equality. If $y = \lambda x$ (or $x = 0$) equality is clear. Conversely, if $|(x,y)| = \|x\|\,\|y\|$ with $x, y \neq 0$, then for $t = \|x\| / \|y\|$ and the α above, $\|x + \alpha t y\|^2 = \|x\|^2 + t^2\|y\|^2 - 2t\,|(x,y)| = 0$, so $x = -\alpha t y$. □

Proof of Proposition 1.6. (N1) is (ii) of Definition 1.2. (N2): $\|\lambda x\|^2 = (\lambda x, \lambda x) = \lambda\overline{\lambda}\,(x,x) = |\lambda|^2\|x\|^2$. (N3): using $\operatorname{Re} w \le |w|$ and (1),

$$\|x+y\|^2 = \|x\|^2 + 2\operatorname{Re}(x,y) + \|y\|^2 \le \|x\|^2 + 2\|x\|\,\|y\| + \|y\|^2 = \big(\|x\| + \|y\|\big)^2. \qquad \square$$

Remark 1.8. Two identities that follow by simply expanding $\|x \pm y\|^2$ are worth remembering:

$$\|x+y\|^2 + \|x-y\|^2 = 2\|x\|^2 + 2\|y\|^2 \qquad \text{(parallelogram law)},$$
$$\|x+y\|^2 = \|x\|^2 + \|y\|^2 \quad \text{if } (x,y) = 0 \qquad \text{(Pythagoras)}.$$

The parallelogram law characterises norms coming from an inner product (Jordan–von Neumann theorem); e.g. $\max_i |x_i|$ on $\mathbb{C}^2$ does not satisfy it.

1.2 Hilbert spaces

Background 9. (Cauchy sequences and completeness)

A sequence (x_n) in a normed space is a *Cauchy sequence* if

$$\forall \varepsilon > 0\ \exists N_\varepsilon\ \forall m, n \ge N_\varepsilon :\ \|x_n - x_m\| < \varepsilon.$$

Every convergent sequence is Cauchy ($\|x_n - x_m\| \le \|x_n - x\| + \|x - x_m\|$). The space is *complete* if conversely every Cauchy sequence converges, i.e. there is x in the space with $\|x_n - x\| \to 0$. A complete normed space is called a *Banach space*. Example: $\mathbb{R}$ and $\mathbb{C}$ are complete, $\mathbb{Q}$ is not ($1,\ 1.4,\ 1.41,\ 1.414, \cdots \to \sqrt{2} \notin \mathbb{Q}$).

Definition 1.10 (Hilbert space)**.** A *Hilbert space* is a linear space H with an inner product $(\cdot,\cdot)$ such that H is complete as a normed space $(H, \|\cdot\|)$, $\|x\| = \sqrt{(x,x)}$: for every Cauchy sequence $(x_n) \subset H$ there exists $x \in H$ with $\|x_n - x\| \xrightarrow{n\to\infty} 0$.

Example 1.11 (An inner product space which is not complete)**.** On $C([0,1])$ consider $(f,g) = \int_0^1 f\overline{g}\,dx$. Let f_n be 0 on $[0, \frac{1}{2}]$, 1 on $[\frac{1}{2} + \frac{1}{n}, 1]$ and linear in between. Then $\|f_n - f_m\|^2 \le \frac{1}{\min(m,n)}$, so (f_n) is Cauchy, but its L^2-limit is the indicator of $(\frac{1}{2}, 1]$, which is not (a.e. equal to) a continuous function. So completeness is a genuine requirement; $L^2([0,1])$ below is the "completion" of this space.

Example 1.12. The following are Hilbert spaces.

(a) $\mathbb{C}^N := \left\{ a = \begin{pmatrix} a_1 \\ \vdots \\ a_N \end{pmatrix} : a_k \in \mathbb{C} \right\}$ with $(a,b)_{\mathbb{C}^N} = \sum_{k=1}^N a_k \overline{b_k}$. Completeness: a Cauchy sequence in $\mathbb{C}^N$ is Cauchy in each coordinate, and $\mathbb{C}$ is complete.

(b) $\ell^2(\mathbb{Z}) := \left\{ \{a_k\}_{k\in\mathbb{Z}} : \sum_{k\in\mathbb{Z}} |a_k|^2 < \infty \right\}$ with $(\{a_k\},\{b_k\})_{\ell^2(\mathbb{Z})} = \sum_{k\in\mathbb{Z}} a_k\overline{b_k}$.

Why this is well defined: $|a_k\overline{b_k}| \le \frac{1}{2}(|a_k|^2 + |b_k|^2)$, so the series converges absolutely; the same inequality shows $\ell^2(\mathbb{Z})$ is closed under addition. *Completeness (sketch):* if $(a^{(n)})$ is Cauchy in ℓ^2, then for each k the numbers $a_k^{(n)}$ form a Cauchy sequence in $\mathbb{C}$, with limit a_k. For $\varepsilon > 0$ and $m, n \ge N_\varepsilon$ and every finite K: $\sum_{|k|\le K} |a_k^{(n)} - a_k^{(m)}|^2 < \varepsilon^2$; let $m \to \infty$, then $K \to \infty$: $\left\| a^{(n)} - a \right\| \le \varepsilon$. Hence $a \in \ell^2$ and $a^{(n)} \to a$.

(c) $L^2(X,\mu)$, see Theorem 1.13 for the notions involved:

$$L^2(X,\mu) := \Big\{ f\colon X \to \mathbb{C} \text{ measurable} : \int_X |f|^2\, d\mu < \infty \Big\} \Big/ \sim, \qquad (f,g) = \int_X f\, \overline{g}\, d\mu,$$

where $f \sim g$ if $f = g$ μ-almost everywhere on X. The quotient is needed for (ii) of Definition 1.2: $\int |f|^2 d\mu = 0$ only gives $f = 0$ a.e., not $f \equiv 0$. Completeness of L^2 is a theorem of measure theory (the *Riesz–Fischer theorem* in its original form).

Obviously $\ell^2(\mathbb{Z}) = L^2(\mathbb{Z}, \#)$, where $\#$ is the *counting measure* on $\mathbb{Z}$ ($\#(A)$ = number of elements of A; the notebook writes $\delta_{\mathbb{Z}}$ – the "discrete measure" $\sum_{n\in\mathbb{Z}} \delta_n$). Then $\int_{\mathbb{Z}} f\, d\# = \sum_{k\in\mathbb{Z}} f(k)$, and only the empty set is a null set.

(d) $S_2(H)$, the Hilbert–Schmidt operators on a Hilbert space H (Theorem 1.14), with $(A,B) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$. For $H = \mathbb{C}^N$ these are all $N \times N$ matrices and $(A,B) = \sum_{i,j} A_{ij}\overline{B_{ij}}$ – the matrix is just regarded as a vector in $\mathbb{C}^{N^2}$.

(e) The *Fock space* (Bargmann–Fock space)

$$\mathcal{F}_2 = \Big\{ f \text{ entire} : \int_{\mathbb{C}} |f(z)|^2 e^{-|z|^2}\, d\lambda_2(z) < \infty \Big\}, \qquad (f,g) = \frac{1}{\pi} \int_{\mathbb{C}} f(z)\, \overline{g(z)}\, e^{-|z|^2}\, d\lambda_2(z),$$

where λ_2 is Lebesgue measure on $\mathbb{C} = \mathbb{R}^2$ and a function is *entire* if it is holomorphic (complex differentiable) on all of $\mathbb{C}$. See Example 1.15.

Background 13. (Measure spaces, measurable functions, a.e.)

A *σ-algebra* on a set X is a family $\mathcal{F}$ of subsets of X containing X and closed under complements and countable unions. A *measure* is a map $\mu\colon \mathcal{F} \to [0,\infty]$ with $\mu(\emptyset) = 0$ and $\mu(\bigsqcup_n A_n) = \sum_n \mu(A_n)$ for disjoint $A_n \in \mathcal{F}$. $(X, \mathcal{F}, \mu)$ is a *measure space*.
The *Borel σ-algebra* $\mathcal{B}(Y)$ of a metric (topological) space Y is the smallest σ-algebra containing all open sets. On $\mathbb{C}$ we use $\mathcal{B}(\mathbb{C}) = \mathcal{B}(\mathbb{R}^2)$. A measure on $\mathcal{B}(X)$ is a *Borel measure*.
$f\colon X \to \mathbb{C}$ is *measurable* if $f^{-1}(B) \in \mathcal{F}$ for every $B \in \mathcal{B}(\mathbb{C})$. A property holds *$\mu$-almost everywhere* (a.e.) if it fails only on a set $N \in \mathcal{F}$ with $\mu(N) = 0$.

Background 14. (Bounded operators, adjoints, Hilbert–Schmidt operators)

An *operator* $A\colon H \to H$ is a linear map. It is *bounded* if $\|A\| := \sup_{\|x\|\le 1} \|Ax\| < \infty$; $\mathcal{B}(H)$ denotes the bounded operators. For $A \in \mathcal{B}(H)$ the *adjoint* $A^* \in \mathcal{B}(H)$ is defined by $(Ax, y) = (x, A^*y)$ for all x, y (for matrices: the conjugate transpose). If $H = \ell^2$, an operator can be thought of as an infinite matrix $((Ae_j, e_i))_{i,j}$ acting on sequences.
$A \in \mathcal{B}(H)$ is *Hilbert–Schmidt* if $\sum_n \|Ae_n\|^2 < \infty$ for an orthonormal basis $\{e_n\}$ (Definition 1.19). The sum does not depend on the choice of the basis: by Parseval (Theorem 1.26) applied twice, for ONBs $\{e_n\}$, $\{f_m\}$,

$$\sum_n \|Ae_n\|^2 = \sum_n \sum_m |(Ae_n, f_m)|^2 = \sum_m \sum_n |(e_n, A^* f_m)|^2 = \sum_m \|A^* f_m\|^2,$$

and the right side does not depend on $\{e_n\}$. We set $\|A\|_{S_2} = \big(\sum_n \|Ae_n\|^2\big)^{1/2}$ and $S_2(H) = \{A \in \mathcal{B}(H) : \sum_n \|Ae_n\|^2 < \infty\}$. The *trace* of an operator is $\operatorname{trace} T = \sum_n (Te_n, e_n)$ (when this makes sense); for $A, B \in S_2(H)$ one has $\operatorname{trace}(B^*A) = \operatorname{trace}(AB^*) = \sum_n (Ae_n, Be_n)$, and this is the inner product of $S_2(H)$.

Example 1.15 (The Fock space)**.** Polynomials, constants, sin, e^{az}, … belong to $\mathcal{F}_2$. Indeed, in polar coordinates $z = re^{i\theta}$, $d\lambda_2 = r\, dr\, d\theta$,

$$\|z^n\|_{\mathcal{F}_2}^2 = \frac{1}{\pi} \int_0^{2\pi} \int_0^\infty r^{2n} e^{-r^2}\, r\, dr\, d\theta = 2 \int_0^\infty r^{2n+1} e^{-r^2} dr \overset{s=r^2}{=} \int_0^\infty s^n e^{-s} ds = n!,$$

and $(z^n, z^m) = 0$ for $n \neq m$ (the θ-integral of $e^{i(n-m)\theta}$ vanishes). So $\{z^n/\sqrt{n!}\}_{n\geq 0}$ is an orthonormal system in $\mathcal{F}_2$ (in fact an orthonormal basis). Since $|\sin z| \leq e^{|z|}$ and $|e^{az}| \leq e^{|a||z|}$, and $\int_{\mathbb{C}} e^{2c|z|-|z|^2} d\lambda_2 < \infty$, also sin and e^{az} belong to $\mathcal{F}_2$. On the other hand $e^{z^2} \notin \mathcal{F}_2$, since $|e^{z^2}|^2 e^{-|z|^2} = e^{|z|^2}$ on the real axis.

1.3 Orthonormal bases

Definition 1.16. Let H be a Hilbert space. A family $\{e_k\}_{k\in I} \subset H$ is an *orthogonal system* if $(e_k, e_j) = 0$ for all $k, j \in I$, $k \neq j$.

Background 17. (Span, closure, density)

$\operatorname{span}\{e_k\}_{k\in I}$ is the set of all *finite* linear combinations $\sum_{k\in F} c_k e_k$, $F \subset I$ finite. The *closure* $\overline{A} = \operatorname{clos}_H A$ of a set $A \subset H$ is the set of all limits of sequences in A; A is *dense* in H if $\overline{A} = H$, i.e. every $h \in H$ can be approximated arbitrarily well by elements of A: $\forall h \in H\ \forall \varepsilon > 0\ \exists a \in A : \|h - a\| < \varepsilon$. $\operatorname{clos}\operatorname{span}\{e_k\} = \overline{\operatorname{span}}\{e_k\}$ is the *closed linear span*: the smallest closed subspace containing all e_k.

Definition 1.18. Let H be a Hilbert space. A family $\{e_k\}_{k\in I}$ is *complete* if $\operatorname{clos}_H \operatorname{span}\{e_k\}_{k\in I} = H$.

Definition 1.19. $\{e_k\}_{k\in I}$ is an *orthonormal basis* (ONB) of H if it is a complete orthogonal system in H with $\|e_k\| = 1$ for all $k \in I$.

Remark 1.20. An ONB is usually *not* a basis in the sense of linear algebra (a Hamel basis): elements of H are in general *infinite* combinations $\sum_k c_k e_k$, converging in norm (Theorem 1.26). E.g. $(1, \frac{1}{2}, \frac{1}{3}, \dots) \in \ell^2(\mathbb{N})$ is not a finite combination of the unit vectors.

Background 21. (Separable spaces)

A metric space X is *separable* if it contains a *countable dense subset*, i.e. a countable $A \subseteq X$ with $\overline{A} = X$. Example: $\mathbb{R}$ is separable since $\mathbb{Q}$ is countable and $\overline{\mathbb{Q}} = \mathbb{R}$; $\mathbb{C}^N$ is separable ($(\mathbb{Q} + i\mathbb{Q})^N$). If A is countable and dense, then for every h there is a sequence $(a_n) \subset A$ with $a_n \to h$ (pick a_n with $\|h - a_n\| < \frac{1}{n}$).

Exercise 1.22. Let H be a separable Hilbert space and $\{e_k\}_{k\in I}$ an ONB in H. Then I is countable.

Solution. For $k \neq j$, by Pythagoras, $\|e_k - e_j\|^2 = \|e_k\|^2 + \|e_j\|^2 = 2$. Hence the open balls $B_k = B(e_k, \frac{\sqrt{2}}{2})$, $k \in I$, are pairwise disjoint (a point in $B_k \cap B_j$ would give $\|e_k - e_j\| < \sqrt{2}$). Let A be a countable dense set. Every ball B_k contains some point $a_k \in A$, and $a_k \neq a_j$ for $k \neq j$ by disjointness. Thus $k \mapsto a_k$ is an injection $I \to A$, so I is countable.

Proposition 1.23. *For a Hilbert space $H \neq \{0\}$ the following are equivalent: (a) H is separable; (b) H has a countable ONB; (c) H has a countable complete system.*

Proof. (a)⇒(c): a countable dense set is a complete system. (c)⇒(b): apply the Gram–Schmidt procedure to the complete sequence (x_n): discard every x_n that lies in $\operatorname{span}\{x_1, \dots, x_{n-1}\}$ and put $e_n = \frac{x_n - \sum_{j<n}(x_n, e_j)e_j}{\|\cdots\|}$. The e_n are orthonormal and have the same span as the x_n, hence they are complete. (b)⇒(a): finite combinations of the e_n with coefficients in $\mathbb{Q} + i\mathbb{Q}$ form a countable set whose closure contains $\operatorname{span}\{e_n\}$, hence equals H. □

From now on, all Hilbert spaces will be separable. To construct a non-separable Hilbert space one needs a "wild" measure, for example the counting measure on an uncountable set: $\ell^2(\mathbb{R}) = L^2(\mathbb{R}, \#)$ consists of $f : \mathbb{R} \to \mathbb{C}$ vanishing outside a countable set with $\sum_x |f(x)|^2 < \infty$, and $\{\mathbb{1}_{\{x\}}\}_{x\in\mathbb{R}}$ is an uncountable ONB.

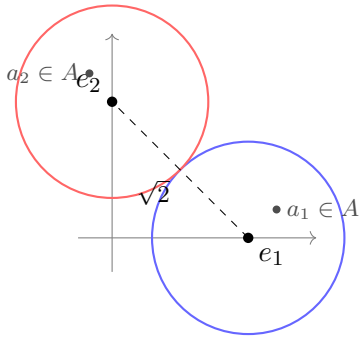


Figure 1: Orthonormal vectors are at distance $\sqrt{2}$, so balls of radius $\sqrt{2}/2$ around them are disjoint; each contains its own point of the dense set A.

1.4 Fourier coefficients and the Riesz–Fischer theorem

Example 1.24 (Observation: the finite-dimensional case)**.** In $\mathbb{C}^N$ the standard basis is $\{e_k\}_{k=1}^N$, $e_k = (0, \dots, 0, \underbrace{1}_{k\text{-th place}}, 0, \dots, 0)^T$, and for all $p \in \mathbb{C}^N$

$$p = p_1 e_1 + \cdots + p_N e_N, \qquad p_k = (p, e_k)_{\mathbb{C}^N}$$

(p_k is the projection coefficient of p onto e_k). This holds for *every* ONB $\{e_k\}$ of $\mathbb{C}^N$: if $p = \sum_j c_j e_j$, then $(p, e_k) = \sum_j c_j (e_j, e_k) = c_k$ by orthonormality. The goal is to prove a similar formula in the infinite-dimensional case.

Definition 1.25. Let H be a Hilbert space, $\{e_k\}_{k=1}^\infty$ an ONB in H and $h \in H$. The number

$$\hat{h}(k) = (h, e_k)$$

is the *k-th Fourier coefficient* of h with respect to $\{e_n\}_{n=1}^\infty$.

Theorem 1.26 (Riesz–Fischer)**.** *Let H be a separable Hilbert space with ONB $\{e_n\}_{n=1}^\infty$. Then for every $h \in H$*

$$h = \sum_{k=1}^{\infty} \hat{h}(k)\, e_k,$$

where the series converges in the norm of H, i.e. $\left\| h - \sum_{k=1}^N \hat{h}(k) e_k \right\|_H \xrightarrow{N\to\infty} 0$. Moreover

$$\|h\|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \qquad \textit{(Parseval's identity).} \tag{3}$$

Remark 1.27. (a) "Converges in the norm" is *not* pointwise convergence: for $H = L^2$ the partial sums need not converge at any given point (this will be the main theme of Section 2).

(b) Names vary in the literature: (3) is usually called *Parseval's identity*; "Riesz–Fischer theorem" often refers to completeness of L^2, or to the converse statement Corollary 1.30.

(c) The statement also holds for non-separable H (with uncountable sums, of which only countably many terms are non-zero), but we will not need this.

We need a simple but important fact first.

Lemma 1.28 (Continuity of the inner product)**.** *If $x_N \to x$ in H, then $(x_N, y) \to (x, y)$ for all $y \in H$, and $\|x_N\| \to \|x\|$.*

Proof. By Cauchy–Schwarz, $|(x_N, y) - (x, y)| = |(x_N - x, y)| \le \|x_N - x\| \, \|y\| \to 0$. By the triangle inequality, $\|x\| \le \|x_N\| + \|x - x_N\|$ and $\|x_N\| \le \|x\| + \|x_N - x\|$, hence $\big|\|x_N\| - \|x\|\big| \le \|x_N - x\| \to 0$. □

Proof of Theorem 1.26. Step 1: Bessel's inequality. We show

$$\sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 \le \|h\|^2 . \tag{4}$$

For this, completeness of $\{e_k\}$ is not needed – it holds for every orthonormal system. Set $h_N = \sum_{k=1}^{N} \hat{h}(k) e_k$. By orthonormality,

$$\Big\|\sum_{k=1}^{N} c_k e_k\Big\|^2 = \Big(\sum_k c_k e_k, \sum_j c_j e_j\Big) = \sum_{k,j} c_k \overline{c_j}\,(e_k, e_j) = \sum_{k=1}^{N} |c_k|^2, \qquad \text{so} \qquad \|h_N\|^2 = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 .$$

Further, by Remark 1.3,

$$(h, h_N) = \Big(h, \sum_{k=1}^{N} \hat{h}(k) e_k\Big) = \sum_{k=1}^{N} \overline{\hat{h}(k)}\,(h, e_k) = \sum_{k=1}^{N} \overline{\hat{h}(k)} \hat{h}(k) = \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \|h_N\|^2 .$$

Summarizing, $(h, h_N) = \|h_N\|^2$, so by Cauchy–Schwarz $\|h_N\|^2 \le \|h\| \, \|h_N\|$, i.e. $\|h_N\| \le \|h\|$ (trivial if $h_N = 0$). Thus $\sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 \le \|h\|^2$ for all N, and letting $N \to \infty$ gives (4).

Step 2: (h_N) is a Cauchy sequence. For $N > M$,

$$\|h_N - h_M\|^2 = \Big\| \sum_{k=M+1}^{N} \hat{h}(k) e_k \Big\|^2 = \sum_{k=M+1}^{N} \left|\hat{h}(k)\right|^2 \le \sum_{k=M+1}^{\infty} \left|\hat{h}(k)\right|^2 \xrightarrow{M\to\infty} 0,$$

since the tails of the convergent series (4) tend to 0. As H is complete, there exists

$$\tilde{h} = \lim_{N\to\infty} h_N = \lim_{N\to\infty} \sum_{k=1}^{N} \hat{h}(k) e_k,$$

in other words $\sum_{k=1}^{\infty} \hat{h}(k) e_k$ converges and $\tilde{h} = \sum_{k=1}^{\infty} \hat{h}(k) e_k$.

Step 3: $\tilde{h} = h$.

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For fixed k and all $N \ge k$ we have $(h_N, e_k) = \sum_{j=1}^{N} \hat{h}(j)\,(e_j, e_k) = \hat{h}(k)$. Hence, by Lemma 1.28 (this is the step that was "left as an exercise"),

$$\big(\tilde{h}, e_k\big) = \big(\lim_{N\to\infty} h_N, e_k\big) = \lim_{N\to\infty} (h_N, e_k) = \hat{h}(k) = (h, e_k)\,,$$

so $\big(h - \tilde{h}, e_k\big) = 0$ for all k: h and $\tilde{h}$ have the same projections onto every e_k. By linearity $\big(h - \tilde{h}, g\big) = 0$ for all $g \in \operatorname{span}\{e_k\}$, and by Lemma 1.28 (in the second slot) for all $g \in \operatorname{clos}\operatorname{span}\{e_k\} = H$ – here completeness of the system is used. Taking $g := h - \tilde{h}$ gives $\big\|h - \tilde{h}\big\|^2 = 0$, i.e. $h = \tilde{h}$.

Step 4: Parseval. Since $h_N \to h$, Lemma 1.28 gives

$$\|h\|^2 = \lim_{N\to\infty} \|h_N\|^2 = \lim_{N\to\infty} \sum_{k=1}^{N} \left|\hat{h}(k)\right|^2 = \sum_{k=1}^{\infty} \left|\hat{h}(k)\right|^2 .$$

□

Corollary 1.29 (Completeness criterion)**.** *An orthonormal sequence $\{e_k\}$ in H is complete if and only if the only $h \in H$ with $(h, e_k) = 0$ for all k is $h = 0$.*

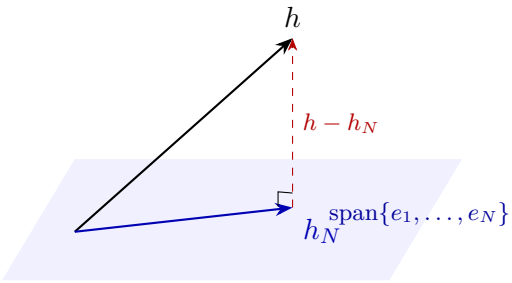


Figure 2: h_N is the orthogonal projection of h onto $\operatorname{span}\{e_1, \dots, e_N\}$: $h - h_N \perp e_k$ for $k \le N$. Pythagoras gives $\|h\|^2 = \|h_N\|^2 + \|h - h_N\|^2 \ge \|h_N\|^2$ – another proof of Bessel's inequality.

Proof. "$\Rightarrow$" is Step 3 above. "$\Leftarrow$": for any h, Steps 1–2 (which do not use completeness) give $\tilde{h} = \sum \hat{h}(k) e_k \in \operatorname{clos}\operatorname{span}\{e_k\}$, and as in Step 3, $\left(h - \tilde{h}, e_k\right) = 0$ for all k. By assumption $h = \tilde{h} \in \operatorname{clos}\operatorname{span}\{e_k\}$. □

Corollary 1.30. *If $\{e_k\}$ is an ONB of H and $(c_k) \in \ell^2(\mathbb{N})$, then $\sum_k c_k e_k$ converges to some $h \in H$ with $\hat{h}(k) = c_k$. Consequently $h \mapsto (\hat{h}(k))_k$ is a bijective linear isometry $H \to \ell^2(\mathbb{N})$: every separable infinite-dimensional Hilbert space "is" ℓ^2.*

Proof. The partial sums are Cauchy exactly as in Step 2; the limit has coefficients c_k as in Step 3. □

Example 1.31 (Euler's formula)**.** $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$

Proof. Let $H = L^2([-\pi, \pi], \mu = \frac{dx}{2\pi})$ and $e_k(x) = e^{ikx}$, $k \in \mathbb{Z}$. Orthonormality is a direct computation: $(e_k, e_j) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(k-j)x} dx = 1$ if $k = j$ and $= \frac{1}{2\pi}\frac{e^{i(k-j)x}}{i(k-j)}\Big|_{-\pi}^{\pi} = 0$ if $k \ne j$. The non-trivial thing is that $\{e_k\}_{k\in\mathbb{Z}}$ is *complete*: "trigonometric functions are dense in L^2". This is proved in Corollary 2.19 below (and nothing proved before it relies on this example). Since $\mathbb{Z}$ is countable, we may enumerate $\{e_k\}_{k\in\mathbb{Z}}$ as a sequence and apply Theorem 1.26.

Consider $f(x) = x$ on $[-\pi, \pi]$. Then $\|f\|_H^2 = \frac{1}{2\pi}\int_{-\pi}^{\pi} x^2 dx = \frac{1}{2\pi}\cdot\frac{2\pi^3}{3} = \frac{\pi^2}{3}$, $\hat{f}(0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\, dx = 0$, and for $k \ne 0$, integrating by parts,

$$\hat{f}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} x\,\overline{e^{ikx}}\,dx = \frac{1}{2\pi}\int_{-\pi}^{\pi} x e^{-ikx} dx = \frac{1}{2\pi}\Big[\frac{xe^{-ikx}}{-ik}\Big]_{-\pi}^{\pi} + \frac{1}{2\pi}\underbrace{\int_{-\pi}^{\pi}\frac{e^{-ikx}}{ik}dx}_{=0}$$

$$= \frac{1}{2\pi}\cdot\frac{\pi e^{-ik\pi} + \pi e^{ik\pi}}{-ik} = \frac{(-1)^k}{-ik} = \frac{i(-1)^k}{k}.$$

Hence $\sum_{k\in\mathbb{Z}}\left|\hat{f}(k)\right|^2 = \sum_{k\ne 0}\frac{1}{k^2} = 2\sum_{k=1}^{\infty}\frac{1}{k^2}$, and Parseval's identity $\|f\|_H^2 = \sum_k |\hat{f}(k)|^2$ – "Pythagoras in infinite dimensions" – gives $\frac{\pi^2}{3} = 2\sum_{k\ge 1}\frac{1}{k^2}$. □

2 Classical Fourier series

2.1 Setting

Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$ be the unit circle and m the Lebesgue (arc-length) measure on $\mathbb{T}$ normalized by $\frac{1}{2\pi}$, so that $m(\mathbb{T}) = 1$. Concretely, $t \mapsto e^{it}$ maps $[-\pi, \pi)$ bijectively onto $\mathbb{T}$ and

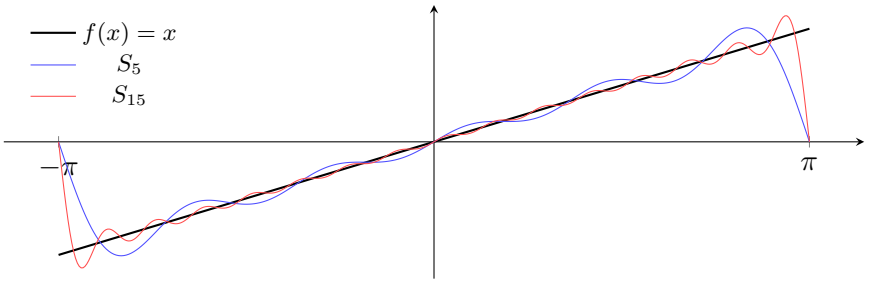


Figure 3: Partial sums $S_N(x) = \sum_{|k|\le N} \hat{f}(k) e^{ikx} = 2\sum_{k=1}^{N} \frac{(-1)^{k+1}}{k} \sin kx$ of $f(x) = x$. They converge to f in L^2; at $\pm\pi$ (the jump of the periodic extension) they are 0, the average of the one-sided limits (cf. Theorem 2.36).

$m(A) = \frac{1}{2\pi}\lambda_1\{t \in [-\pi, \pi) : e^{it} \in A\}$. We work in

$$H = L^2(\mathbb{T}, m), \qquad (f, g) = \underbrace{\int_{\mathbb{T}} f\,\overline{g}\,dm}_{\text{general measure theory}} = \underbrace{\frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})\,\overline{g(e^{it})}\,dt}_{\text{parametrization}}.$$

Functions on $\mathbb{T}$ are the same as 2π-periodic functions on $\mathbb{R}$, via $g(t) = f(e^{it})$.

Exercise 2.1. $\{z^n\}_{n\in\mathbb{Z}}$ is an ONB in $L^2(\mathbb{T}, m)$.

Solution. Orthonormality: $\left(z^n, z^k\right) = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{i(n-k)t}dt = \delta_{nk}$, as in Example 1.31. *Completeness* needs two facts.

(1) Trigonometric polynomials $\sum_{|k|\le N} c_k z^k$ are dense in $C(\mathbb{T})$ with the sup-norm. This follows from the Stone–Weierstrass theorem (Theorem 2.2), since they form an algebra containing 1, closed under conjugation ($\overline{z} = z^{-1}$ on $\mathbb{T}$) and separating points (already z does). We will also get a self-contained proof from Fejér's theorem (Corollary 2.18).

(2) $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$ (Theorem 2.3).

Given $f \in L^2$ and $\varepsilon > 0$, choose $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$ and a trigonometric polynomial p with $\max_{\mathbb{T}} |g - p| < \varepsilon$. Since $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} \le \max_{\mathbb{T}} |g - p|$, so $\|f - p\|_{L^2} < 2\varepsilon$. Hence $\operatorname{clos}\operatorname{span}\{z^n\} = L^2(\mathbb{T}, m)$.

Background 2. (Weierstrass and Stone–Weierstrass theorems)

Weierstrass approximation theorem. Every continuous $f\colon [a, b] \to \mathbb{C}$ is a uniform limit of polynomials. *Trigonometric version:* every $f \in C(\mathbb{T})$ is a uniform limit of trigonometric polynomials.
Stone–Weierstrass theorem. Let K be a compact metric space and $\mathcal{A} \subset C(K)$ a subalgebra (closed under sums, products and scalar multiples) which contains the constants, separates points ($\forall x \ne y\ \exists a \in \mathcal{A} : a(x) \ne a(y)$) and is closed under complex conjugation. Then $\mathcal{A}$ is dense in $C(K)$ with respect to $\|f\|_\infty = \max_K |f|$.

Background 3. (Continuous functions are dense in L^p)

If μ is a finite Borel measure on a compact metric space K and $1 \le p < \infty$, then $C(K)$ is dense in $L^p(K, \mu)$. (Idea: simple functions are dense; by regularity of μ an indicator $\mathbb{1}_A$ is approximated by $\mathbb{1}_F$ with $F \subset A$ closed, and $\mathbb{1}_F$ by the continuous functions $\max(0, 1 - n\operatorname{dist}(x, F))$.)

Definition 2.4. For $f \in L^2(\mathbb{T}, m)$ (or even $f \in L^1(\mathbb{T}, m)$) the Fourier coefficients are

$$\hat{f}(k) = \left(f, z^k\right) = \int_{\mathbb{T}} f(z)\,\overline{z}^k\,dm(z) = \int_{\mathbb{T}} f(z)\,z^{-k}\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it})e^{-ikt}dt, \qquad k \in \mathbb{Z},$$

using $\overline{z} = z^{-1}$ on $\mathbb{T}$. The *Fourier series* of f is $\sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ and its symmetric partial sums are $S_N(f,z) = \sum_{k=-N}^{N} \hat{f}(k) z^k$.

By the Riesz–Fischer theorem, for all $f \in L^2(\mathbb{T}, m)$ we have $f = \sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ where the series converges in $L^2(\mathbb{T}, m)$, i.e.

$$\int_{\mathbb{T}} \Big| f - \sum_{k=-N}^{N} \hat{f}(k) z^k \Big|^2 dm \xrightarrow{N\to\infty} 0. \tag{$**$}$$

(Here we use that $\mathbb{Z}$ can be enumerated as $0, 1, -1, 2, -2, \dots$, so the symmetric partial sums are a subsequence of the partial sums of Theorem 1.26.)

But ($**$) is not what one would naively expect: we are interested in *pointwise* convergence. In other words, we would like to have $f(z) = \sum_{k\in\mathbb{Z}} \hat{f}(k) z^k$ a.e. on $\mathbb{T}$ for "good" f. This is true, but very hard to prove:

Theorem 2.5 (Carleson 1966, Hunt 1968)**.** *If $f \in L^p(\mathbb{T}, m)$ for some $p \in (1, \infty]$, then $S_N(f,z) \xrightarrow{N\to\infty} f(z)$ for m-almost every $z \in \mathbb{T}$.*

For $p = 1$ this fails: Kolmogorov (1923) constructed $f \in L^1(\mathbb{T})$ whose Fourier series diverges almost everywhere (even everywhere). We will prove much simpler results, starting with convergence in a weaker sense.

Definition 2.6 (Cesàro convergence)**.** A sequence $(S_n)_{n\geq 0}$ *converges in the Cesàro sense* to L if its arithmetic means converge: $\sigma_N := \frac{1}{N+1} \sum_{n=0}^{N} S_n \xrightarrow{N\to\infty} L$. A series converges in the Cesàro sense if the sequence of its partial sums does.

Example 2.7. $S_n = -1, 1, -1, 1, \dots$ (the partial sums of $-1 + 2 - 2 + 2 - \dots$) does not converge, but $\sigma_N \to 0$: the means are 0 or $-\frac{1}{N+1}$.

Proposition 2.8 (Cesàro summation is consistent)**.** *If $S_n \to L$, then also $\sigma_N \to L$.*

Proof. Let $\varepsilon > 0$ and $|S_n - L| < \varepsilon$ for $n > n_0$. Then $|\sigma_N - L| \leq \frac{1}{N+1} \sum_{n=0}^{n_0} |S_n - L| + \frac{N-n_0}{N+1}\varepsilon$, and the first term tends to 0 as $N \to \infty$. So $\limsup |\sigma_N - L| \leq \varepsilon$ for every ε. □

Theorem 2.9 (Fejér)**.** *Let $f \in C(\mathbb{T})$ (a continuous function on the unit circle). Then the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to f in the Cesàro sense everywhere on $\mathbb{T}$. In other words, if $S_N(z) = \sum_{k=-N}^{N} \hat{f}(k) z^k$, then*

$$\frac{1}{N+1} \sum_{n=0}^{N} S_n(z) \longrightarrow f(z) \qquad \text{for every } z \in \mathbb{T}.$$

The proof is given in Section 2.4; first we write the Cesàro means as an integral.

2.2 The Fejér kernel

Lemma 2.10. $\displaystyle \frac{1}{N+1} \sum_{n=0}^{N} S_n(z) = \int_{\mathbb{T}} f(\xi)\, F_N(\xi, z)\, dm(\xi)$, *where*

$$F_N(\xi, z) := \frac{|1 + \overline{\xi} z + \cdots + (\overline{\xi} z)^N|^2}{N+1} = \frac{1}{N+1} \Big| \sum_{s=0}^{N} (\overline{\xi} z)^s \Big|^2$$

is the Fejér kernel.

Proof. Since all sums are finite, we may interchange them with the integral:

$$\frac{1}{N+1}\sum_{n=0}^{N} S_n(z) = \frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n} \hat f(k) z^k = \frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n}\int_{\mathbb{T}} f(\xi)\overline{\xi}^k \, dm(\xi)\, z^k$$
$$= \int_{\mathbb{T}} f(\xi)\Big(\frac{1}{N+1}\sum_{n=0}^{N}\sum_{k=-n}^{n} \overline{\xi}^k z^k\Big) dm(\xi).$$

So the lemma reduces to the identity, for $w = \overline{\xi} z \in \mathbb{T}$,

$$\sum_{n=0}^{N}\sum_{k=-n}^{n} w^k = \Big|\sum_{s=0}^{N} w^s\Big|^2. \tag{5}$$

Direct proof. Using $\overline{w} = w^{-1}$,

$$\Big|\sum_{s=0}^{N} w^s\Big|^2 = \Big(\sum_{s=0}^{N} w^s\Big)\Big(\sum_{r=0}^{N} w^{-r}\Big) = \sum_{k=-N}^{N} \#\{(s,r) : s-r=k\}\, w^k = \sum_{k=-N}^{N} (N+1-|k|)\, w^k,$$

and on the left of (5) the power w^k appears once for every n with $|k| \le n \le N$, i.e. $N+1-|k|$ times.

Proof by induction (as in the lecture). For $N=0$ both sides equal 1. Write $P_N = \sum_{s=0}^{N} w^s$ and suppose (5) holds for N. The left side for $N+1$ is $\sum_{k=-(N+1)}^{N+1} w^k + |P_N|^2$. Using $|a+b|^2 = |a|^2 + |b|^2 + 2\,\mathrm{Re}(a\bar b)$ with $a = P_N$, $b = w^{N+1}$:

$$|P_{N+1}|^2 = |P_N|^2 + \underbrace{|w^{N+1}|^2}_{=1} + 2\,\mathrm{Re}\big(P_N \overline{w}^{N+1}\big), \qquad P_N\overline{w}^{N+1} = \sum_{s=0}^{N} w^{s-N-1} = \overline{w} + \overline{w}^2 + \cdots + \overline{w}^{N+1},$$

so $|P_{N+1}|^2 = |P_N|^2 + 1 + \sum_{j=1}^{N+1}(w^j + \overline{w}^j) = |P_N|^2 + \sum_{k=-(N+1)}^{N+1} w^k$, as needed. □

Remark 2.11 (Closed form; Dirichlet kernel). Write $z = e^{i\theta}$, $\xi = e^{it}$, $u = \theta - t$, so $w = \overline{\xi} z = e^{iu}$. Summing the geometric series and using $|1 - e^{ia}| = 2|\sin\frac{a}{2}|$,

$$F_N(\xi, z) = K_N(u) := \frac{1}{N+1}\frac{|1-e^{i(N+1)u}|^2}{|1-e^{iu}|^2} = \frac{1}{N+1}\left(\frac{\sin\frac{(N+1)u}{2}}{\sin\frac{u}{2}}\right)^2.$$

Similarly $S_N(f, e^{i\theta}) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(\theta - t)\, dt$ with the *Dirichlet kernel*

$$D_N(u) = \sum_{k=-N}^{N} e^{iku} = \frac{\sin\big((N+\frac12)u\big)}{\sin\frac{u}{2}}, \qquad \text{and} \qquad K_N = \frac{1}{N+1}\sum_{n=0}^{N} D_n.$$

The decisive difference: $K_N \ge 0$ while D_N changes sign, and $\frac{1}{2\pi}\int |D_N| \to \infty$ (Section 2.5). Both kernels are even functions of u.

2.3 Approximate identities

Definition 2.12 (Approximate identity)**.** Let X be a metric space with metric ρ, μ a Borel measure on X and $x_0 \in X$. An *approximate identity with center at* x_0 is a family of measurable functions p_n on X such that

(i) $p_n(x) \ge 0$ for all $x \in X$;

(ii) $\int_X p_n(x)\, d\mu(x) = 1$ for all n;

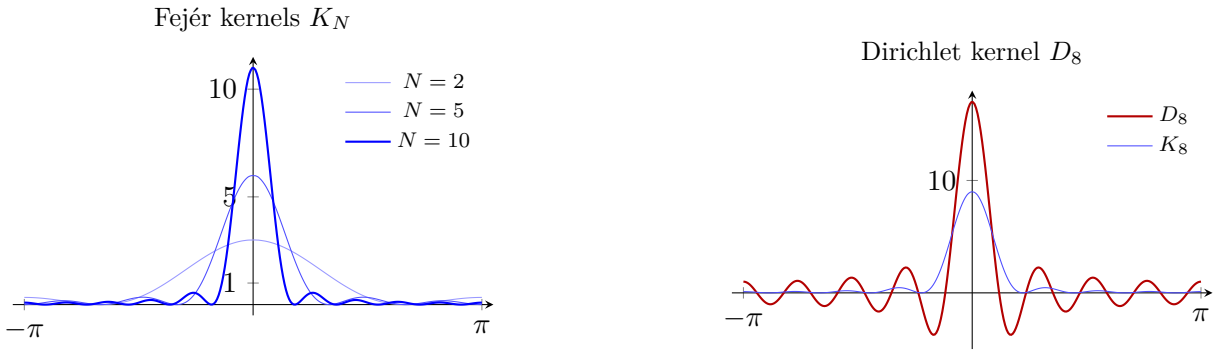


Figure 4: Left: $K_N \geq 0$, $\frac{1}{2\pi}\int K_N = 1$, and the mass concentrates at 0 – an approximate identity. Right: the Dirichlet kernel oscillates and takes negative values; its absolute integral grows like $\log N$.

(iii) $\int_{X\setminus B(x_0,\delta)} p_n(x)\,d\mu(x) \xrightarrow{n\to\infty} 0$ for every $\delta > 0$, where $B(x_0,\delta) = \{y \in X : \rho(y,x_0) < \delta\}$.

Remark 2.13. (a) In the lecture (iii) was first stated as $\sup_{x\in X\setminus B(x_0,\delta)} p_n(x) \to 0$; the integral version above (suggested in the lecture as "simpler") is weaker when $\mu(X) < \infty$, since $\int_{X\setminus B} p_n\,d\mu \leq \mu(X)\sup_{X\setminus B} p_n$, and it is exactly what the proof below needs. For the Fejér kernel both hold.

(b) Whether the ball is open or closed does not matter.

(c) Intuition: the mass 1 of $p_n\,d\mu$ concentrates at x_0. In the language of distributions, $p_n \to \delta_{x_0}$ (Dirac delta): $\int f\,p_n\,d\mu \to \int f\,d\delta_{x_0} = f(x_0)$ – that is the next lemma.

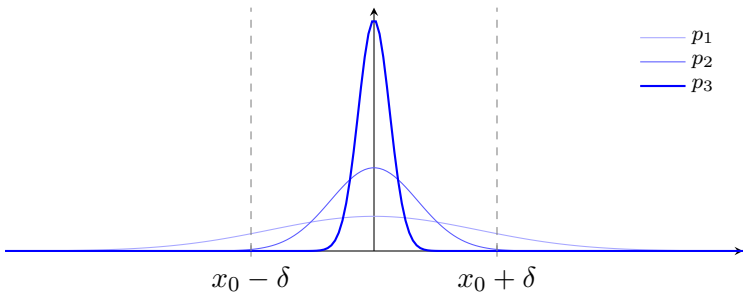


Figure 5: An approximate identity: non-negative, total mass 1, and the mass outside $B(x_0,\delta)$ tends to 0.

Lemma 2.14. *Let (p_n) be an approximate identity with center at x_0 and let f be a bounded measurable function on X which is continuous at x_0 (e.g. f bounded and continuous). Then*

$$\int_X f(x)\,p_n(x)\,d\mu(x) \xrightarrow{n\to\infty} f(x_0).$$

Proof. By (ii), $f(x_0) = \int_X f(x_0)p_n\,d\mu$, so using (i),

$$\begin{aligned}\Big|\int_X f\,p_n\,d\mu - f(x_0)\Big| &= \Big|\int_X (f(x)-f(x_0))p_n(x)\,d\mu(x)\Big| \\ &\leq \int_{B(x_0,\delta)} |f(x)-f(x_0)|\,p_n(x)\,d\mu + \int_{X\setminus B(x_0,\delta)} |f(x)-f(x_0)|\,p_n(x)\,d\mu \\ &\leq \sup_{x\in B(x_0,\delta)} |f(x)-f(x_0)| \cdot \underbrace{\int_{B(x_0,\delta)} p_n\,d\mu}_{\leq 1 \text{ by (i),(ii)}} + 2\sup_{y\in X}|f(y)| \cdot \int_{X\setminus B(x_0,\delta)} p_n\,d\mu.\end{aligned}$$

Let $\varepsilon > 0$. By continuity at x_0 choose $\delta > 0$ with $|f(x) - f(x_0)| < \frac{\varepsilon}{2}$ on $B(x_0, \delta)$; then, by (iii), choose N such that the second summand is $< \frac{\varepsilon}{2}$ for $n \geq N$. □

We will see that $F_N(\xi, z)$ is an approximate identity (in ξ) at the point z.

Remark 2.15 (Dense subsets, once more). "A subset $D = \{x_1, x_2, x_3, \dots\}$ is dense in a Hilbert space H if for every $h \in H$ and every $\varepsilon > 0$ there exists some $x_n \in D$ such that $\|h - x_n\|_H < \varepsilon$." This is the definition from Theorem 1.17 specialised to countable sets; such H is separable (Theorem 1.21).

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2.4 Proof of Fejér's theorem

Recap. $H = L^2(\mathbb{T}, m)$, $\mathbb{T} = \{|z| = 1\}$, ONB $\{z^k\}_{k \in \mathbb{Z}}$,

$$S_n(f, z) := \sum_{k=-n}^{n} \hat{f}(k) z^k, \qquad \sigma_N(f, z) := \frac{1}{N+1} \sum_{n=0}^{N} S_n(f, z).$$

Theorem 2.16 (Fejér)**.** *For every $f \in C(\mathbb{T})$ we have $\sigma_N(f, z) \xrightarrow{N \to \infty} f(z)$ for every $z \in \mathbb{T}$. Equivalently, the Fourier series $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k$ converges to $f(z)$ in the Cesàro sense at every point $z \in \mathbb{T}$.*

Proof. By Lemma 2.10, $\sigma_N(f, z) = \int_{\mathbb{T}} f(\xi) F_N(\xi, z)\, dm(\xi)$ with $F_N(\xi, z) = \frac{1}{N+1} \left|\sum_{s=0}^{N} (\overline{\xi} z)^s\right|^2$. Fix z. We check that $(F_N(\cdot, z))_N$ is an approximate identity in the variable ξ with center at z, on $X = \mathbb{T}$ with $\rho(\xi, z) = |\xi - z|$ and $\mu = m$:

(i) $F_N(\xi, z) \geq 0$ for all $\xi \in \mathbb{T}$. ✓

(ii) $\int_{\mathbb{T}} F_N(\xi, z)\, dm(\xi) = 1$. Indeed, $\xi \mapsto 1 + \overline{\xi} z + \dots + \overline{\xi}^N z^N = \sum_{s=0}^{N} z^s\, \xi^{-s}$ is a finite combination of the orthonormal functions ξ^{-s}, so by Parseval (or just orthonormality)

$$\int_{\mathbb{T}} \left|1 + \overline{\xi} z + \dots + \overline{\xi}^N z^N\right|^2 dm(\xi) = 1 + |z|^2 + |z^2|^2 + \dots + |z^N|^2 \overset{|z|=1}{=} N + 1. ✓$$

(iii) For $\xi \in \mathbb{T} \setminus B(z, \delta)$ we have $|1 - \overline{\xi} z| = |\overline{\xi}|\, |\xi - z| = |\xi - z| \geq \delta$, and $|1 - \overline{\xi}^{N+1} z^{N+1}| \leq 2$. Summing the geometric series,

$$\int_{\mathbb{T} \setminus B(z,\delta)} F_N(\xi, z)\, dm(\xi) \leq \underbrace{m(\mathbb{T} \setminus B(z, \delta))}_{\leq m(\mathbb{T}) = 1} \cdot \sup_{\xi \in \mathbb{T} \setminus B(z,\delta)} \frac{1}{N+1} \left|\frac{1 - \overline{\xi}^{N+1} z^{N+1}}{1 - \overline{\xi} z}\right|^2$$
$$\leq \frac{2^2}{\delta^2 (N+1)} \xrightarrow[\delta \text{ fixed}]{N \to \infty} 0.$$

So $F_N(\cdot, z)$ is an approximate identity with center at z, and f is bounded and continuous on the compact set $\mathbb{T}$. By Lemma 2.14, $\sigma_N(f, z) \to f(z)$, for every z. □

Remark 2.17 (Uniform convergence). In fact $\sigma_N(f, \cdot) \rightrightarrows f$ uniformly on $\mathbb{T}$, i.e. $\max_{z \in \mathbb{T}} |\sigma_N(f, z) - f(z)| \to 0$, $N \to \infty$, because all estimates were uniform in z. *Check:* f is uniformly continuous on the compact set $\mathbb{T}$, so for $\varepsilon > 0$ there is δ *not depending on* z with $|f(\xi) - f(z)| < \frac{\varepsilon}{2}$ whenever $|\xi - z| < \delta$. The proof of Lemma 2.14 then gives

$$|\sigma_N(f, z) - f(z)| \leq \frac{\varepsilon}{2} + 2\, \|f\|_\infty \cdot \frac{4}{\delta^2 (N+1)} \qquad \text{for all } z \in \mathbb{T},$$

and the right side is $< \varepsilon$ for N large, independently of z.

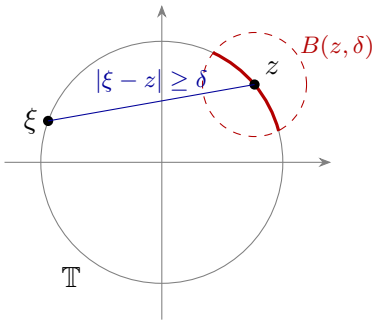


Figure 6: For ξ outside the arc $B(z,\delta)\cap\mathbb{T}$ the denominator $|1-\overline{\xi}z| = |\xi - z|$ stays $\geq \delta$, so $F_N(\xi,z) \leq \frac{4}{\delta^2(N+1)}$ there.

Corollary 2.18 (Weierstrass' trigonometric approximation theorem)**.** *Trigonometric polynomials are dense in* $(C(\mathbb{T}), \|\cdot\|_\infty)$.

Proof. $\sigma_N(f,\cdot) = \sum_{|k|\leq N}(1-\frac{|k|}{N+1})\hat{f}(k)z^k$ is a trigonometric polynomial and $\sigma_N(f,\cdot)\to f$ uniformly by Remark 2.17. □

Corollary 2.19. $\{z^n\}_{n\in\mathbb{Z}}$ *is an ONB of* $L^2(\mathbb{T},m)$*;*
equivalently $\{e^{ikx}\}_{k\in\mathbb{Z}}$ *is an ONB of* $L^2([-\pi,\pi],\frac{dx}{2\pi})$*. In particular Example 1.31 is justified.*

Proof. As in the solution of Exercise 2.1, with Corollary 2.18 in place of Stone–Weierstrass. □

Corollary 2.20. *If* $f\in C(\mathbb{T})$ *and* $S_N(f,z)$ *converges at some point* z*, then the limit is* $f(z)$ *(by Proposition 2.8 and Fejér's theorem).*

Remark 2.21 (Continuity at one point suffices). If f is bounded and measurable on $\mathbb{T}$ (not necessarily in $C(\mathbb{T})$) and $f(\xi)\to f(z_0)$ as $\xi\to z_0$, then $\sigma_N(f,z_0)\to f(z_0)$. *Check:* Lemma 2.14 only uses boundedness of f and continuity at the center $x_0 = z_0$.

Remark 2.22 (Jump discontinuities – homework). Let f be bounded and measurable, and let $z_0 = e^{it_0}$ be such that the one-sided limits

$$f(z_0-0) := \lim_{\varepsilon\to 0,\ \varepsilon>0} f(e^{i(t_0-\varepsilon)}), \qquad f(z_0+0) := \lim_{\varepsilon\to 0,\ \varepsilon>0} f(e^{i(t_0+\varepsilon)})$$

exist (f need not be continuous at z_0). Then

$$\sigma_N(f,z_0) \longrightarrow \frac{f(z_0-0)+f(z_0+0)}{2} \qquad (N\to\infty).$$

This generalizes Fejér's theorem (if f is continuous at z_0 both limits equal $f(z_0)$).

Solution. By Remark 2.11, $\sigma_N(f,z_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{i(t_0+u)})K_N(u)\,du$ (substitute $t = t_0+u$ and use that K_N is even and 2π-periodic). Since K_N is even and $\frac{1}{2\pi}\int_{-\pi}^{\pi}K_N = 1$, we have $\frac{1}{2\pi}\int_0^{\pi}K_N = \frac{1}{2\pi}\int_{-\pi}^{0}K_N = \frac{1}{2}$. Hence, with $L_\pm = f(z_0\pm 0)$,

$$\sigma_N(f,z_0) - \frac{L_-+L_+}{2} = \frac{1}{2\pi}\int_0^{\pi}(f(e^{i(t_0+u)})-L_+)K_N(u)\,du + \frac{1}{2\pi}\int_0^{\pi}(f(e^{i(t_0-u)})-L_-)K_N(u)\,du.$$

Each term tends to 0 by the argument of Lemma 2.14: given ε, choose δ such that $|f(e^{i(t_0+u)})-L_+|<\varepsilon$ for $0<u<\delta$; the part $\int_0^\delta$ is $\leq\frac{\varepsilon}{2}$, and the part $\int_\delta^\pi$ is $\leq 2\|f\|_\infty\cdot\frac{1}{2\pi}\int_{\delta\leq|u|\leq\pi}K_N\to 0$, because $K_N(u)\leq\frac{1}{(N+1)\sin^2(\delta/2)}$ there.

2.5 Divergence: the du Bois-Reymond theorem

Theorem 2.23 (du Bois-Reymond)**.** *There exists a function $f \in C(\mathbb{T})$ such that $S_N(f,z)$ does not converge at some point $z \in \mathbb{T}$.*

So for continuous functions the Cesàro means in Fejér's theorem are really necessary. The proof is based on a classical result of functional analysis.

Background 24. (Banach spaces, bounded operators and functionals)

A *Banach space* is a complete normed linear space (Theorem 1.9). Example: $C(\mathbb{T})$ with $\|f\| = \max_{z\in\mathbb{T}} |f(z)|$ – a uniform limit of continuous functions is continuous, so Cauchy sequences converge in $C(\mathbb{T})$. A linear map $T\colon X \to Y$ between normed spaces is *bounded* if $\|T\| := \sup_{\|x\|\le 1} \|Tx\| < \infty$; for linear maps, bounded $\iff$ continuous. A bounded linear map $X \to \mathbb{C}$ is a *bounded linear functional*.

Theorem 2.25 (Banach–Steinhaus, uniform boundedness principle)**.** *Let X be a Banach space, Y a normed linear space and $\{T_\alpha\}_{\alpha\in A}$ a family of bounded linear operators $X \to Y$ such that $\sup_{\alpha\in A} \|T_\alpha x\| < \infty$ for every $x \in X$. Then*

$$\sup_{\alpha\in A} \|T_\alpha\| = \sup_{\alpha\in A} \sup_{x\in X,\ \|x\|\le 1} \|T_\alpha x\| < \infty.$$

The proof was skipped in the lecture as a basic fact from functional analysis; here is the standard argument.

Background 26. (Baire category theorem)

If a complete metric space is a countable union of closed sets, then at least one of them contains an open ball.

Proof of Theorem 2.25 (sketch). The sets $E_n = \{x \in X : \sup_\alpha \|T_\alpha x\| \le n\}$ are closed (intersections of closed sets, by continuity of each T_α) and $\bigcup_n E_n = X$ by assumption. By Baire, some E_n contains a ball $B(x_0, r)$. For $\|x\| \le 1$, both x_0 and $x_0 + rx$ lie in E_n, so $\|T_\alpha x\| = \frac{1}{r}\|T_\alpha(x_0 + rx) - T_\alpha x_0\| \le \frac{2n}{r}$ for all α. □

Proof of Theorem 2.23. Let $X := C(\mathbb{T})$ with $\|f\| = \max_{z\in\mathbb{T}} |f(z)|$ (a Banach space), $Y := \mathbb{C}$ and

$$\varphi_N(f) := S_N(f,1) = \sum_{k=-N}^{N} \hat f(k), \qquad f \in C(\mathbb{T}).$$

φ_N is linear, $\varphi_N(\alpha f + \beta g) = \alpha\varphi_N(f) + \beta\varphi_N(g)$, because $\hat f(k) = \left(f, z^k\right)_{L^2}$ is linear in f. It is bounded: $|\hat f(k)| = \left|\int_{\mathbb{T}} f\,\overline{z}^k\,dm\right| \le \max_{\mathbb{T}} |f| \cdot m(\mathbb{T}) = \|f\|$, hence $\|\varphi_N\| \le 2N + 1$.

Suppose $S_N(f,1)$ converged (to a finite limit) for every $f \in C(\mathbb{T})$. Then $\sup_N |\varphi_N(f)| < \infty$ for every f, and Banach–Steinhaus gives $\sup_{N\ge 0} \|\varphi_N\| < \infty$. We show that this is not the case, which proves that there is $f \in C(\mathbb{T})$ with $\sup_N |S_N(f,1)| = \infty$; in particular $S_N(f,1)$ diverges. (Rotating f moves the bad point 1 to any prescribed $z \in \mathbb{T}$.)

Computing $\|\varphi_N\|$. Since $\sum_{k=-N}^{N} \overline{\xi}^k = D_N(-t) = D_N(t)$ for $\xi = e^{it}$ (Remark 2.11),

$$\varphi_N(f) = \sum_{k=-N}^{N} \int_{\mathbb{T}} f(\xi)\overline{\xi}^k\,dm(\xi) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{it}) D_N(t)\,dt.$$

Note D_N is real. In the lecture one takes f_N with $|f_N| = 1$ and $f_N \cdot D_N = |D_N|$, i.e. $f_N = \operatorname{sgn} D_N$; this function is not continuous (it jumps at the zeros of D_N), so we approximate it: $g_j = \frac{D_N}{|D_N| + 1/j}$

is continuous, $|g_j| \le 1$, and $g_j \to \operatorname{sgn} D_N$ wherever $D_N \neq 0$, i.e. except at finitely many points. By dominated convergence,

$$\|\varphi_N\| \ge \lim_{j\to\infty} |\varphi_N(g_j)| = \frac{1}{2\pi}\int_{-\pi}^{\pi} |D_N(t)|\,dt =: L_N \qquad \text{(the } \textit{Lebesgue constants}\text{)}.$$

So it suffices to prove $L_N \to \infty$. Multiplying by $|e^{iNt}| = 1$ and summing the geometric series,

$$\begin{aligned} L_N &= \frac{1}{2\pi}\int_{-\pi}^{\pi} |e^{-iNt} + \cdots + e^{iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} |1 + e^{it} + \cdots + e^{2iNt}|dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|1 - e^{i(2N+1)t}|}{|1 - e^{it}|}dt \\ &\ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\operatorname{Im}(1 - e^{i(2N+1)t})|}{|1 - e^{it}|}dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|1 - e^{it}|}dt \ge \frac{1}{2\pi}\int_{-\pi}^{\pi} \frac{|\sin((2N+1)t)|}{|t|}dt, \end{aligned}$$

using $|a| \ge |\operatorname{Im} a|$, $\operatorname{Im}(e^{ix}) = \sin x$, and $|1 - e^{it}| \le |t|$ (a chord is shorter than the arc, Fig. 7). By evenness, the substitution $u = (2N+1)t$, and $|\sin u| \ge \sin^2 u = \frac{1-\cos 2u}{2}$,

$$\begin{aligned} L_N &\ge \frac{1}{\pi}\int_0^{(2N+1)\pi} \frac{|\sin u|}{u}\,du \ge \frac{1}{2\pi}\int_1^{(2N+1)\pi} \frac{1 - \cos(2u)}{u}\,du \\ &= \frac{1}{2\pi}\Big(\log((2N+1)\pi) - \int_1^{(2N+1)\pi} \frac{\cos(2u)}{u}\,du\Big). \end{aligned}$$

The last integral stays bounded: integrating by parts, $\int_1^R \frac{\cos 2u}{u}\,du = \left[\frac{\sin 2u}{2u}\right]_1^R + \int_1^R \frac{\sin 2u}{2u^2}du$, which is at most $\frac{1}{2} + \frac{1}{2} + \frac{1}{2}$ in absolute value (it converges to $\int_1^\infty \frac{\cos 2u}{u}\,du$). Hence $L_N \ge \frac{1}{2\pi}(\log((2N+1)\pi) - \frac{3}{2}) \to \infty$. □

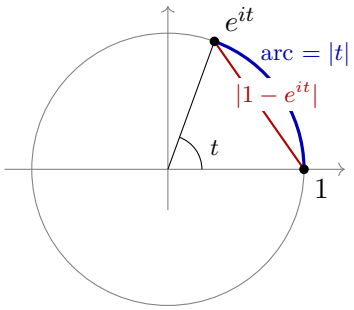


Figure 7: $|1 - e^{it}| = 2|\sin\frac{t}{2}| \le |t|$: the chord is shorter than the arc.

2.6 Functions of bounded variation

Definition 2.27. Let $f\colon \mathbb{T} \to \mathbb{C}$. We say that f has *bounded variation*, $f \in BV(\mathbb{T})$, if

$$V_{\mathbb{T}}(f) := \sup\Big\{\sum_{j=1}^{n} |f(z_{j+1}) - f(z_j)|\Big\} < \infty,$$

the supremum taken over all finite partitions $z_1, \dots, z_n$ of $\mathbb{T}$ (points ordered counter-clockwise, $z_{n+1} := z_1$).

Definition 2.28. Let $g\colon [a,b] \to \mathbb{C}$. Then $g \in BV([a,b])$ if

$$V_a^b(g) := \sup_{a=t_1<t_2<\cdots<t_n=b} \sum_{k=1}^{n-1} |g(t_{k+1}) - g(t_k)| < \infty.$$

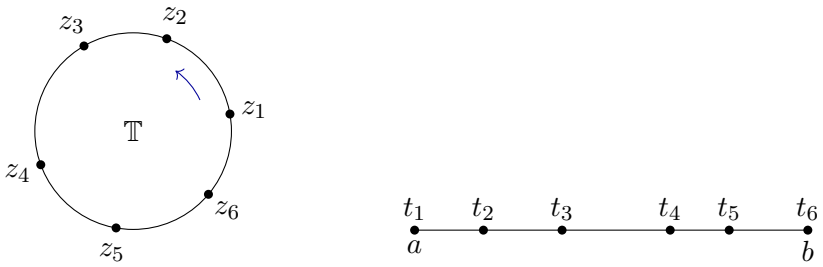


Figure 8: Partitions of $\mathbb{T}$ and of $[a,b]$ used in the definition of the total variation.

Remark 2.29. Let $g(t) = f(e^{it})$. Then $g \in BV([0,2\pi]) \iff f \in BV(\mathbb{T})$, and in fact $V_{\mathbb{T}}(f) = V_0^{2\pi}(g)$. Indeed, adding a point to a partition can only increase the sum (triangle inequality), so in both suprema we may assume the point $1 = e^{i0}$ is included; and partitions of $\mathbb{T}$ containing 1 correspond exactly to partitions $0 = t_1 < \cdots < t_{n+1} = 2\pi$ of $[0,2\pi]$, with the same sums (as $g(2\pi) = g(0)$). Also, BV functions are bounded: $|g(t)| \le |g(a)| + V_a^b(g)$.

Theorem 2.30 (Jordan decomposition)**.** $g \in BV([a,b]) \iff$ *there are bounded nondecreasing functions* $g_1, g_2, g_3, g_4 \colon [a,b] \to \mathbb{R}$ *with*

$$g = g_1 - g_2 + i(g_3 - g_4).$$

Proof. "⇐": a nondecreasing h has $\sum_k |h(t_{k+1}) - h(t_k)| = h(b) - h(a)$ for every partition, and sums of BV functions are BV. "⇒": it suffices to treat real g (apply it to $\operatorname{Re} g$, $\operatorname{Im} g$, whose variation is $\le V(g)$). Let $V(x) = V_a^x(g)$. Then V is nondecreasing, and for $x < y$, $g(y) - g(x) \le V_x^y(g) = V(y) - V(x)$, so $V - g$ is nondecreasing as well; $g = V - (V - g)$. □

Background 31. (Complex measures and Lebesgue–Stieltjes measures)

A *complex measure* on $\mathcal{B}(\mathbb{R})$ is a map $\mu \colon \mathcal{B}(\mathbb{R}) \to \mathbb{C}$ with $\mu(\bigsqcup A_n) = \sum \mu(A_n)$ (absolutely convergent) for disjoint A_n. Its *total variation* is the positive measure $|\mu|(A) = \sup \sum_j |\mu(A_j)|$ (sup over finite partitions of A into Borel sets); it is always finite: $|\mu|(\mathbb{R}) < \infty$, and $\left|\int h\, d\mu\right| \le \int |h|\, d|\mu|$.
Lebesgue–Stieltjes: for every nondecreasing, right-continuous, bounded F on $[a,b]$ there is a unique finite Borel measure μ_F on $(a,b]$ with $\mu_F((a,x]) = F(x) - F(a)$.

Theorem 2.32. $(g \in BV([a,b])$ *and* g *is continuous from the right)* $\iff g(x) = g(a) + \int_{(a,x]} d\mu(s)$ *for all* x*, where* μ *is a complex valued measure on* $(a,b]$ *(of finite total variation).*

Idea. "⇒": write $g = g_1 - g_2 + i(g_3 - g_4)$ as in Theorem 2.30; one can choose the g_j right-continuous when g is, and then $\mu = \mu_{g_1} - \mu_{g_2} + i(\mu_{g_3} - \mu_{g_4})$. "⇐": $\sum_k |g(t_{k+1}) - g(t_k)| = \sum_k |\mu((t_k, t_{k+1}])| \le |\mu|((a,b])$, and right-continuity follows from continuity of μ along decreasing sets $(a, x + \frac{1}{n}] \downarrow (a,x]$. □

Example 2.33 (The Cantor staircase)**.** Let $C = \{\sum_{n\ge1} a_n 3^{-n} : a_n \in \{0,2\}\} \subset [0,1]$ be the Cantor set (remove middle thirds repeatedly); $\lambda(C) = 0$. The *Cantor function* $F(\sum a_n 3^{-n}) = \sum \frac{a_n}{2} 2^{-n}$ on C, extended to be constant on each removed interval, is continuous, nondecreasing, with $F(0) = 0$, $F(1) = 1$. So $F \in BV([0,1])$ and $F(x) = \mu((0,x])$ for the *Cantor measure* μ. Note $F' = 0$ a.e., yet F is not constant: μ has no atoms but lives on the null set C (it is *singular continuous*). Hence μ in Theorem 2.32 cannot in general be replaced by $g'(s)\, ds$.

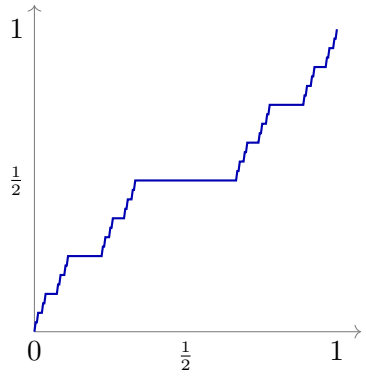


Figure 9: The Cantor staircase (5th stage of the construction): a continuous BV function whose derivative is 0 almost everywhere.

Example 2.34. If $g \in C^1([a,b])$, then $g \in BV([a,b])$:

$$\sum_k |g(t_{k+1}) - g(t_k)| \le c \sum_k |t_{k+1} - t_k| = c\,(b-a) < \infty, \qquad c = \max_{[a,b]} |g'|,$$

by the inequality $|g(x) - g(y)| = \left|\int_y^x g'\right| \le \max_{[a,b]} |g'| \cdot |x-y|$ (for complex g use this integral form, the mean value theorem itself fails for complex valued functions). Here μ in Theorem 2.32 is $d\mu = g'(s)\,ds$: the derivative of such a function "is" a finite measure.

Remark 2.35. If $g \in BV([a,b])$, then for all $x_0 \in [a,b]$ the one-sided limits

$$g(x_0 - 0) = \lim_{x \to x_0,\, x < x_0} g(x), \qquad g(x_0 + 0) = \lim_{x \to x_0,\, x > x_0} g(x)$$

exist. Moreover g has at most countably many discontinuities.

Proof. By Theorem 2.30, $g = g_1 - g_2 + i(g_3 - g_4)$, and for bounded monotone functions $g_k(x_0 \pm 0)$ exist (e.g. $g_k(x_0 - 0) = \sup_{x<x_0} g_k(x)$ for nondecreasing g_k). A nondecreasing h is discontinuous at x iff its jump $h(x+0) - h(x-0) > 0$; since any finitely many jumps sum to at most $h(b) - h(a)$, there are at most $n(h(b) - h(a))$ jumps $> \frac{1}{n}$, so countably many jumps altogether. The discontinuities of g lie in the union of those of $g_1, \dots, g_4$. □

2.7 Hardy's theorem (Dirichlet–Jordan test)

Theorem 2.36 (Hardy)**.** *Let $f \in BV(\mathbb{T})$. Then for every $z \in \mathbb{T}$*

$$\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k := \lim_{N \to \infty} S_N(f, z) = \frac{f(z-0) + f(z+0)}{2}.$$

In particular, if $f \in BV(\mathbb{T})$ is continuous at $z \in \mathbb{T}$, then $\sum_{k=-\infty}^{\infty} \hat{f}(k) z^k = f(z)$.

(This is classically known as the Dirichlet–Jordan theorem; Hardy's contribution is the Tauberian argument below.) The strategy: by Fejér the series is Cesàro summable; if in addition the terms are $O(\frac{1}{k})$, Cesàro convergence upgrades to ordinary convergence. Results of the form "summable in a weak sense + a size condition on the terms ⇒ convergent" are called *Tauberian theorems.*

Lemma 2.37 (Hardy's Tauberian theorem)**.** *Assume that $\sum_{n=1}^{\infty} a_n$ converges in the Cesàro sense and $|a_k| \le \frac{C}{k}$ for all $k \ge 1$ and some $C > 0$. Then $\sum_{k=1}^{\infty} a_k$ converges in the usual sense, to the same sum.*

Proof. Notation:

$$S_n := \sum_{k=1}^{n} a_k, \qquad \sigma_n := \frac{1}{n}\sum_{k=1}^{n} S_k \to \sigma, \qquad \sigma_{n,j} := \frac{S_{n+1}+\cdots+S_{n+j}}{j}, \qquad l_n(\varepsilon) = [\varepsilon n]$$

for $\varepsilon \in (0,1]$, $n \geq 1$, where $[\,\cdot\,]$ is the integer part. We only consider $n \geq 1/\varepsilon$, so that $j = l_n(\varepsilon) \geq 1$; note $\frac{\varepsilon n}{2} \leq j \leq \varepsilon n$. The proof has two steps:

- Step 1: $\lim_{n\to\infty} |\sigma_n - \sigma_{n,l_n(\varepsilon)}| = 0$;
- Step 2: $\limsup_{n\to\infty} |S_n - \sigma_{n,l_n(\varepsilon)}| \leq C\varepsilon$ for every ε.

Step 1 (uses only Cesàro convergence). Since $S_1 + \cdots + S_m = m\sigma_m$,

$$\sigma_{n,j} = \frac{(S_1+\cdots+S_{n+j}) - (S_1+\cdots+S_n)}{j} = \sigma_{n+j}\frac{n+j}{j} - \sigma_n\frac{n}{j} = (\sigma_{n+j}-\sigma_n)\frac{n+j}{j} + \sigma_n.$$

Thus, with $j = l_n(\varepsilon)$,

$$|\sigma_{n,j} - \sigma_n| = |\sigma_{n+j} - \sigma_n| \cdot \frac{n+j}{j} \leq \underbrace{|\sigma_{n+l_n(\varepsilon)} - \sigma_n|}_{\to 0} \cdot \underbrace{\frac{n+\varepsilon n}{\varepsilon n/2}}_{=2\frac{1+\varepsilon}{\varepsilon},\ \text{indep. of } n} \longrightarrow 0,$$

because (σ_m) converges, so $\sigma_{n+j} - \sigma_n \to 0$ for any $j = j(n) \geq 0$.

Step 2 (uses only the size condition). Since $S_n - S_{n+i} = -(a_{n+1} + \cdots + a_{n+i})$,

$$\begin{aligned}\Big|S_n - \frac{S_{n+1}+\cdots+S_{n+j}}{j}\Big| &= \frac{1}{j}\Big|\sum_{i=1}^{j}(S_n - S_{n+i})\Big| \leq \frac{1}{j}\sum_{i=1}^{j}(|a_{n+1}| + \cdots + |a_{n+i}|) \\ &\leq |a_{n+1}| + \cdots + |a_{n+j}| \leq C\Big(\frac{1}{n+1} + \cdots + \frac{1}{n+j}\Big) \\ &\leq \frac{Cj}{n+1} \leq C\,\frac{\varepsilon n}{n+1} \leq C\varepsilon.\end{aligned}$$

Taking the lim sup we obtain Step 2.

Conclusion. $S_n \to \sigma$ iff $\lim |S_n - \sigma| = 0$. By the triangle inequality,

$$\limsup_n |S_n - \sigma| \leq \underbrace{\limsup_n |S_n - \sigma_{n,l_n(\varepsilon)}|}_{\leq C\varepsilon \text{ by Step 2}} + \underbrace{\limsup_n |\sigma_{n,l_n(\varepsilon)} - \sigma_n|}_{=0 \text{ by Step 1}} + \underbrace{\limsup_n |\sigma_n - \sigma|}_{=0 \text{ by Cesàro convergence}},$$

so $\limsup_n |S_n - \sigma| \leq C\varepsilon$ for every $\varepsilon > 0$, i.e. $\lim |S_n - \sigma| = 0$. □

Remark 2.38 (Indexing). Fejér's means are $\frac{1}{N+1}\sum_{n=0}^{N} S_n$ for a series $\sum_{n\geq 0} a_n$, while Lemma 2.37 uses $\frac{1}{n}\sum_{k=1}^{n} S_k$ for $\sum_{n\geq 1} a_n$. This is only a shift of index: putting $b_n = a_{n-1}$ $(n \geq 1)$, the partial sums and means of $\sum b_n$ are S_{n-1} and $\frac{1}{n}\sum_{k=0}^{n-1} S_k$, and $|a_n| \leq \frac{C}{n+1}$ for $n \geq 0$ becomes $|b_n| \leq \frac{C}{n}$.

Lemma 2.39. *If $f \in BV(\mathbb{T})$, then $|\hat{f}(k)| \leq \frac{C}{|k|+1}$ for all $k \in \mathbb{Z}$, with C depending on f only.*

Proof (lecture version, assuming f continuous from the right). Let $g(t) = f(e^{it})$ for $t \in [0, 2\pi]$; then $g \in BV([0,2\pi])$ (remark after the definition of BV). Assume g is continuous from the right (an extra condition). Then, by Theorem 2.32, $g(t) = g(0) + \int_{(0,t]} d\mu(s)$ for a finite complex measure μ. For $k \neq 0$,

$$\begin{aligned}\hat{f}(k) &= \int_{\mathbb{T}} f(\xi)\bar{\xi}^k\, dm(\xi) = \frac{1}{2\pi}\int_0^{2\pi} g(t)e^{-ikt}dt = \frac{1}{2\pi}\int_0^{2\pi}\Big(g(0) + \int_{(0,t]} d\mu(s)\Big)e^{-ikt}dt \\ &= \underbrace{\frac{g(0)}{2\pi}\int_0^{2\pi} e^{-ikt}dt}_{=0 \text{ for } k\neq 0} + \frac{1}{2\pi}\int_0^{2\pi}\int_{(0,2\pi]} \chi_{(0,t]}(s)\, e^{-ikt}\, d\mu(s)\, dt \\ &\overset{\text{Fubini}}{=} \frac{1}{2\pi}\int_{(0,2\pi]}\Big(\int_s^{2\pi} e^{-ikt}dt\Big)d\mu(s),\end{aligned}$$

where Fubini applies since the integrand is bounded and $|\mu|$, dt are finite. Now $\int_s^{2\pi} e^{-ikt}dt = \frac{e^{-ikt}}{-ik}\Big|_s^{2\pi} = \frac{e^{-iks}-1}{ik}$ has modulus $\le \frac{2}{|k|}$, so

$$|\hat f(k)| \le \frac{1}{2\pi}\cdot\frac{2}{|k|}\cdot|\mu|((0,2\pi]) = O\Big(\frac{1}{|k|}\Big).$$

Together with $|\hat f(0)| \le \sup|f|$ this gives the claim, since $\frac{1}{|k|} \le \frac{2}{|k|+1}$ for $k \neq 0$. "In general this assumption is not important, just a longer proof" – here is a short proof without it. □

Proof (general). Extend $g(t) = f(e^{it})$ 2π-periodically; let $V = V_{\mathbb{T}}(f)$. Fix $k \neq 0$ and $h = \frac{\pi}{|k|}$, so $e^{-ik(t+h)} = -e^{-ikt}$. Shifting the variable of integration by h (periodicity), $\hat f(k) = \frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ik(t+h)}dt = -\frac{1}{2\pi}\int_0^{2\pi} g(t+h)e^{-ikt}dt$. Averaging with the original formula,

$$|\hat f(k)| = \Big|\frac{1}{4\pi}\int_0^{2\pi}(g(t)-g(t+h))e^{-ikt}dt\Big| \le \frac{1}{4\pi}\int_0^{2\pi}|g(t+h)-g(t)|\,dt.$$

Since $2\pi = 2|k|\cdot h$, split $[0,2\pi)$ into $2|k|$ intervals of length h:

$$\int_0^{2\pi}|g(t+h)-g(t)|\,dt = \int_0^h\sum_{j=0}^{2|k|-1}|g(s+(j+1)h)-g(s+jh)|\,ds \le h\,V,$$

because for each s the points $e^{i(s+jh)}$ form a partition of $\mathbb{T}$. Hence $|\hat f(k)| \le \frac{hV}{4\pi} = \frac{V}{4|k|}$. □

Proof of Theorem 2.36. Fix $z \in \mathbb{T}$. A BV function is bounded and has one-sided limits $f(z\pm 0)$ at every $z \in \mathbb{T}$ (Remark 2.35). By Fejér's theorem in the form of Remark 2.22, $\sum_{k=-\infty}^{\infty}\hat f(k)z^k$ converges in the Cesàro sense to $\frac{f(z-0)+f(z+0)}{2}$.

Reduction to a continuous-from-the-right f (lecture). Changing f at its points of discontinuity, e.g. replacing it by $\tilde f(e^{it}) = g(t+0)$, does not change $\sum\hat f(k)z^k$ nor $\frac{f(z-0)+f(z+0)}{2}$: the set of discontinuities is countable (Remark 2.35), hence of measure 0, so the integrals $\hat f(k) = \int f\bar\xi^k\,dm$ are unchanged; and one-sided limits are not affected by the values at single points. Also $V_{\mathbb{T}}(\tilde f) \le V_{\mathbb{T}}(f)$ (partition sums of $\tilde f$ are limits of partition sums of f). With the general proof of Lemma 2.39 this reduction is not even needed.

Tauberian step. By Lemma 2.39, $|\hat f(k)| \le \frac{C}{|k|+1}$ for all k. Group the terms symmetrically: $a_0 = \hat f(0)$ and $a_n = \hat f(n)z^n + \hat f(-n)z^{-n}$ for $n \ge 1$. Then $\sum_{n=0}^N a_n = S_N(f,z)$ and $|a_n| \le \frac{2C}{n+1}$. The series $\sum a_n$ converges in the Cesàro sense, so by Lemma 2.37 (with Remark 2.38) it converges in the usual sense to the same value:

$$\sum_{k\in\mathbb{Z}}\hat f(k)z^k = \lim_{N\to\infty}S_N(f,z) = \frac{f(z-0)+f(z+0)}{2}.$$ □

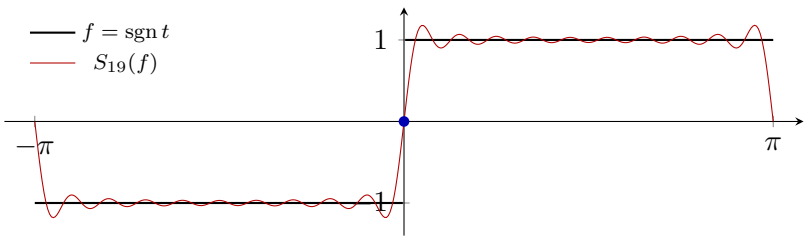


Figure 10: Hardy's theorem for the BV function $f(e^{it}) = \operatorname{sgn} t$: at the jump $t = 0$ every partial sum equals $0 = \frac{f(1-0)+f(1+0)}{2}$; at points of continuity $S_N(f) \to f$.

3 Exercises

The problem sheets of the course, with detailed solutions. The sheets write $\langle \cdot, \cdot \rangle$ for the inner product and χ_A for the indicator function of a set A; we keep their notation here, so $\langle h, e \rangle = (h, e)$ in the notation of the lectures.

3.1 List 1 (13. 10. 2026)

Problem 1. Let H be a (complex, separable) Hilbert space.

(a) Let $(e_i)_{i \in I}$ be an ONS (orthonormal system). Using separability, prove that I is at most countable. *Hint.* Show that $\|e_i - e_j\| = \sqrt{2}$ for $i \neq j$, and use a countable dense subset of H.

(b) Let $e_1, \dots, e_N$ be orthonormal, and put

$$V_N = \operatorname{span}\{e_1, \dots, e_N\}, \qquad P_N h = \sum_{k=1}^{N} \langle h, e_k \rangle e_k.$$

Verify that $P_N h$ is the orthogonal projection of h onto V_N, and then use the Pythagorean theorem to deduce Bessel's inequality: $\sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2 \leq \|h\|^2$.

(c) Let $(e_n)_{n=1}^{\infty}$ be an ONS. Prove that the following are equivalent:

(i) $(e_n)_{n=1}^{\infty}$ is an ONB;
(ii) $\langle h, e_k \rangle = 0$ for all $k \geq 1$ implies $h = 0$;
(iii) for every $h \in H$, $h = \sum_{k=1}^{\infty} \langle h, e_k \rangle e_k$ in H;
(iv) for every $h \in H$, $\|h\|^2 = \sum_{k=1}^{\infty} |\langle h, e_k \rangle|^2$.

Hint. For (iv)⇒(i), you may use the Pythagorean theorem.

Solution. **(a)** This is Exercise 1.22; we repeat the argument for an arbitrary ONS (completeness is not needed). For $i \neq j$, by orthogonality (Pythagoras),

$$\|e_i - e_j\|^2 = \langle e_i - e_j, e_i - e_j \rangle = \|e_i\|^2 - \langle e_i, e_j \rangle - \langle e_j, e_i \rangle + \|e_j\|^2 = 1 - 0 - 0 + 1 = 2.$$

Let $D \subset H$ be countable and dense (it exists since H is separable, Theorem 1.21). For each $i \in I$ the open ball $B_i = B(e_i, \frac{\sqrt{2}}{2})$ contains some point of D (density); choose one and call it d_i. If $d_i = d_j$ for $i \neq j$, then $\|e_i - e_j\| \leq \|e_i - d_i\| + \|d_j - e_j\| < \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2} = \sqrt{2}$, a contradiction. So $i \mapsto d_i$ is an injective map $I \to D$, and I is at most countable.

(b) *Orthogonal projection.* Recall that the orthogonal projection of h onto a subspace V is the (unique) vector $v \in V$ with $h - v \perp V$, i.e. $\langle h - v, w \rangle = 0$ for all $w \in V$. Uniqueness: if v, v' both have this property then $v - v' \in V$ and $v - v' = (h - v') - (h - v) \perp V$, so $\langle v - v', v - v' \rangle = 0$.

Clearly $P_N h \in V_N$. For $j \leq N$, by orthonormality,

$$\langle h - P_N h, e_j \rangle = \langle h, e_j \rangle - \sum_{k=1}^{N} \langle h, e_k \rangle \langle e_k, e_j \rangle = \langle h, e_j \rangle - \langle h, e_j \rangle = 0,$$

and by linearity (and conjugate linearity in the second slot) $\langle h - P_N h, w \rangle = 0$ for every $w \in V_N$. Hence $P_N h$ is the orthogonal projection of h onto V_N. It is also the point of V_N closest to h: for $v \in V_N$, $P_N h - v \in V_N$, so by Pythagoras $\|h - v\|^2 = \|h - P_N h\|^2 + \|P_N h - v\|^2 \geq \|h - P_N h\|^2$.

Bessel. Since $h = P_N h + (h - P_N h)$ with $P_N h \perp h - P_N h$, the Pythagorean theorem gives

$$\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2 \geq \|P_N h\|^2 = \Big\|\sum_{k=1}^{N} \langle h, e_k\rangle e_k\Big\|^2 = \sum_{k=1}^{N} |\langle h, e_k\rangle|^2,$$

the last equality again by orthonormality. Now let $e_1, e_2, \ldots$ be an infinite ONS: the inequality holds for every N, the partial sums of $\sum_k |\langle h, e_k\rangle|^2$ are nondecreasing and bounded by $\|h\|^2$, so the series converges and $\sum_{k=1}^{\infty} |\langle h, e_k\rangle|^2 \leq \|h\|^2$. (Compare with the proof of Theorem 1.26, Step 1, where the same inequality was obtained from Cauchy–Schwarz.)

(c) Throughout, $P_N h = \sum_{k=1}^{N} \langle h, e_k\rangle e_k$ as in (b), and we use $\|h\|^2 = \|P_N h\|^2 + \|h - P_N h\|^2$ and $\|P_N h\|^2 = \sum_{k \leq N} |\langle h, e_k\rangle|^2$.

(i)⇒(ii). Let $\langle h, e_k\rangle = 0$ for all k. Then $\langle h, w\rangle = 0$ for all $w \in \operatorname{span}\{e_k\}$. If $w_n \to w$, then $\langle h, w\rangle = \lim \langle h, w_n\rangle = 0$ (continuity of the inner product, Lemma 1.28); so $\langle h, w\rangle = 0$ for all $w \in \operatorname{clos}\operatorname{span}\{e_k\} = H$. Take $w = h$: $\|h\|^2 = 0$, so $h = 0$.

(ii)⇒(iii). Let $h \in H$. For $N > M$, $\|P_N h - P_M h\|^2 = \sum_{k=M+1}^{N} |\langle h, e_k\rangle|^2$, which tends to 0 as $M \to \infty$ by Bessel's inequality (tails of a convergent series). So $(P_N h)$ is Cauchy, and since H is complete it has a limit $\tilde{h} = \sum_{k=1}^{\infty} \langle h, e_k\rangle e_k$. For fixed k and $N \geq k$, $\langle P_N h, e_k\rangle = \langle h, e_k\rangle$; letting $N \to \infty$, $\langle \tilde{h}, e_k\rangle = \langle h, e_k\rangle$. Thus $\langle h - \tilde{h}, e_k\rangle = 0$ for all k, and (ii) gives $h = \tilde{h}$.

(iii)⇒(iv). By (iii), $P_N h \to h$, hence $\|P_N h\| \to \|h\|$ (Lemma 1.28), i.e. $\sum_{k=1}^{N} |\langle h, e_k\rangle|^2 = \|P_N h\|^2 \to \|h\|^2$.

(iv)⇒(i). The system is orthonormal by assumption; we must show completeness. By Pythagoras and (iv),

$$\|h - P_N h\|^2 = \|h\|^2 - \|P_N h\|^2 = \|h\|^2 - \sum_{k=1}^{N} |\langle h, e_k\rangle|^2 \xrightarrow{N\to\infty} 0.$$

So $h = \lim P_N h$ is a limit of elements of $\operatorname{span}\{e_k\}$, i.e. $h \in \operatorname{clos}\operatorname{span}\{e_k\}$. As h was arbitrary, $\operatorname{clos}\operatorname{span}\{e_k\} = H$ and (e_n) is an ONB.

Remark 3.1. Problem 1(c) is a reformulation of the Riesz–Fischer theorem (Theorem 1.26) together with Corollary 1.29: (i)⇒(iii),(iv) is Theorem 1.26, and (ii) is the practical test for completeness.

Problem 2. Let (X, μ) be a finite measure space.

(a) Prove that $L^2(X,\mu) \subseteq L^1(X,\mu)$ and that $\|f\|_{L^1} \leq \mu(X)^{1/2} \|f\|_{L^2}$. In particular, explain why the Fourier coefficients of functions in $L^2(\mathbb{T}, m)$ are well defined, where m is the normalized arc length measure defined in the lectures, given by $\int_{\mathbb{T}} f(z)\, dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{ix})\, dx$.

(b) For $\alpha \geq 0$ consider $f_\alpha(x) = |x|^{-\alpha}$, $x \in (-1,1) \setminus \{0\}$. Determine the values of α for which $f_\alpha \in L^1(-1,1)$.

(c) Let $f \in L^2(X,\mu)$ and set $f_M = f \cdot \chi_{\{|f| \leq M\}}$, $M > 0$. Prove that $f_M \to f$ in $L^2(X,\mu)$ as $M \to \infty$. *Hint.* Apply DCT (the dominated convergence theorem) to $|f - f_M|^2$.

(d) On $(0,1)$ with Lebesgue measure, consider $f_n = \sqrt{n}\, \chi_{(0,1/n)}$. Show that $f_n \to 0$ almost everywhere but not in L^2. Explain why DCT cannot be used here to conclude L^2 convergence.

Background 2. (L^1 and the convergence theorems)

$L^1(X,\mu)$ is the space of (classes of a.e. equal) measurable f with $\|f\|_{L^1} = \int_X |f|\, d\mu < \infty$; for such f the integral $\int_X f\, d\mu \in \mathbb{C}$ is defined and $\left|\int f\, d\mu\right| \leq \int |f|\, d\mu$.
Monotone convergence theorem (MCT). If $0 \leq g_1 \leq g_2 \leq \ldots$ are measurable and $g_n \to g$ pointwise, then $\int g_n\, d\mu \to \int g\, d\mu$ (possibly $= \infty$).
Dominated convergence theorem (DCT). If $g_n \to g$ μ-a.e. and there is an integrable $G \geq 0$ (independent of n) with $|g_n| \leq G$ a.e. for all n, then $g \in L^1$ and $\int |g_n - g|\, d\mu \to 0$; in particular

$\int g_n \to \int g$.

Solution. **(a)** Let $f \in L^2(X,\mu)$. The constant function 1 is in $L^2(X,\mu)$ since $\int_X 1^2\,d\mu = \mu(X) < \infty$. By Cauchy–Schwarz in L^2 (Theorem 1.7) applied to $|f|$ and 1,

$$\|f\|_{L^1} = \int_X |f| \cdot 1\,d\mu = \langle |f|, 1\rangle_{L^2} \le \|f\|_{L^2}\,\|1\|_{L^2} = \mu(X)^{1/2}\,\|f\|_{L^2} < \infty.$$

So $f \in L^1$. (Functions equal a.e. have the same L^1 and L^2 norms, so this is a statement about classes, and the inclusion map $L^2 \to L^1$ is a bounded linear map.)

Fourier coefficients. For $f \in L^2(\mathbb{T}, m)$ and $k \in \mathbb{Z}$, the function $z \mapsto f(z)\overline{z}^k$ is measurable and $|f(z)\overline{z}^k| = |f(z)|$ since $|z| = 1$. As $m(\mathbb{T}) = 1$, $\int_{\mathbb{T}} |f\overline{z}^k|\,dm = \|f\|_{L^1} \le \|f\|_{L^2} < \infty$, so

$$\hat{f}(k) = \int_{\mathbb{T}} f(z)\overline{z}^k\,dm(z) = \frac{1}{2\pi}\int_{-\pi}^{\pi} f(e^{ix})e^{-ikx}dx$$

is the integral of an integrable function, hence a well defined complex number, with $|\hat{f}(k)| \le \|f\|_{L^1} \le \|f\|_{L^2}$. It does not depend on the representative of f (changing f on a null set does not change the integral). Note also $\hat{f}(k) = \langle f, z^k\rangle_{L^2}$, which is finite by Cauchy–Schwarz directly. (Since $m(\mathbb{T}) = 1$ the coefficients make sense even for $f \in L^1(\mathbb{T}, m)$, a larger space by (b).)

(b) $f_\alpha \ge 0$ is even, so $\int_{-1}^1 f_\alpha = 2\int_0^1 x^{-\alpha}dx$. By MCT, $\int_0^1 x^{-\alpha}dx = \lim_{\varepsilon\downarrow 0}\int_\varepsilon^1 x^{-\alpha}dx$, and

$$\int_\varepsilon^1 x^{-\alpha}dx = \begin{cases} \dfrac{1-\varepsilon^{1-\alpha}}{1-\alpha} \xrightarrow{\varepsilon\to 0} \dfrac{1}{1-\alpha}, & \alpha < 1, \\ -\log\varepsilon \xrightarrow{\varepsilon\to 0} \infty, & \alpha = 1, \\ \dfrac{\varepsilon^{1-\alpha}-1}{\alpha-1} \xrightarrow{\varepsilon\to 0} \infty, & \alpha > 1. \end{cases}$$

Hence $f_\alpha \in L^1(-1,1)$ if and only if $0 \le \alpha < 1$, and then $\|f_\alpha\|_{L^1} = \frac{2}{1-\alpha}$.

Comment. Since $f_\alpha^2 = f_{2\alpha}$, we get $f_\alpha \in L^2(-1,1) \iff \alpha < \frac{1}{2}$. So for $\frac{1}{2} \le \alpha < 1$, $f_\alpha \in L^1 \setminus L^2$: the inclusion in (a) is strict. On spaces of infinite measure there is no inclusion at all, e.g. $\frac{1}{1+|x|} \in L^2(\mathbb{R}) \setminus L^1(\mathbb{R})$.

(c) f_M is measurable ($\{|f| \le M\}$ is a measurable set) and

$$|f - f_M|^2 = |f|^2\,\chi_{\{|f|>M\}}.$$

Pointwise convergence. Since $\int |f|^2 d\mu < \infty$, $|f(x)| < \infty$ for a.e. x. For such x, as soon as $M \ge |f(x)|$ we have $\chi_{\{|f|>M\}}(x) = 0$; hence $|f - f_M|^2(x) \to 0$ as $M \to \infty$, for a.e. x.

Domination. $|f - f_M|^2 \le |f|^2$, and $|f|^2 \in L^1$ because $f \in L^2$.

By DCT (along any sequence $M_n \to \infty$), $\|f - f_M\|_{L^2}^2 = \int_X |f - f_M|^2\,d\mu \to 0$. Since $M \mapsto \|f - f_M\|_{L^2}$ is nonincreasing, convergence along sequences implies convergence as $M \to \infty$ through all reals. (Finiteness of μ is not needed here.)

Meaning. Bounded functions are dense in L^2: every f is approximated by its truncations f_M, $|f_M| \le M$.

(d) *Pointwise convergence.* Fix $x \in (0,1)$. For $n > \frac{1}{x}$ we have $x \notin (0, \frac{1}{n})$, so $f_n(x) = 0$. Hence $f_n(x) \to 0$ for *every* $x \in (0,1)$, in particular almost everywhere.

No L^2 convergence. $\|f_n\|_{L^2}^2 = \int_0^{1/n}(\sqrt{n})^2 dx = n \cdot \frac{1}{n} = 1$, so $\|f_n - 0\|_{L^2} = 1 \not\to 0$. Moreover (f_n) has no L^2-limit at all: if $f_n \to g$ in L^2, a subsequence would converge to g a.e. (a standard fact), so $g = 0$ a.e., contradicting $\|f_n\| = 1$.

Why DCT fails. To apply DCT to $|f_n - 0|^2 = n\chi_{(0,1/n)}$ we would need an integrable G with $n\chi_{(0,1/n)} \le G$ for all n. The smallest candidate is $G = \sup_n n\chi_{(0,1/n)}$. For $x \in [\frac{1}{n+1}, \frac{1}{n})$ we have $x \in (0, \frac{1}{k})$ exactly for $k \le n$, so $G(x) = n > \frac{1}{x} - 1$. Thus

$$\int_0^1 G\,dx \ge \int_0^1 \Big(\frac{1}{x} - 1\Big)dx = \infty,$$

and no integrable dominating function exists. The mass $\int f_n^2 = 1$ does not disappear; it concentrates in the shrinking interval $(0, \frac{1}{n})$ (it "escapes to a point"), which pointwise convergence cannot see.

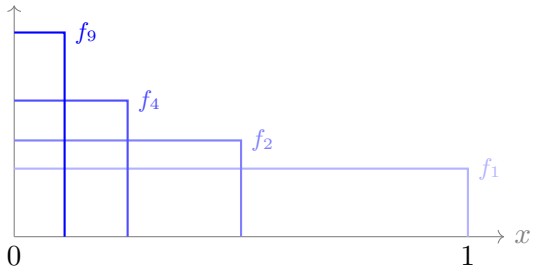


Figure 11: $f_n = \sqrt{n}\,\chi_{(0,1/n)}$: higher and narrower bumps, each with $\int f_n^2 = 1$.

Problem 3. Let $\mathbb{T} = \{z \in \mathbb{C} : |z| = 1\}$.

(a) Show that $\{z^k : k \in \mathbb{Z}\}$ is an ONS in $L^2(\mathbb{T}, m)$, where m is the normalized arc length measure defined above.

(b) Let $I \subset \mathbb{T}$ be a closed arc of positive length strictly less than 2π, i.e. $I = \{e^{it} : x_0 \le t \le x_1\}$, $0 < x_1 - x_0 < 2\pi$. Construct continuous functions φ_ε on $\mathbb{T}$ such that $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon \to \chi_I$ in $L^2(\mathbb{T}, m)$ $(\varepsilon \to 0^+)$. *Hint.* We only need to modify χ_I near the endpoints of I.

(c) (optional) Prove that for every measurable set $A \subseteq \mathbb{T}$ and every $\varepsilon > 0$ there is a finite union of arcs $U \subseteq \mathbb{T}$ such that $m(A \triangle U) < \varepsilon$. In particular, $\|\chi_U - \chi_A\|_{L^2}^2 < \varepsilon$.

(d) Deduce that $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) Let $\mathcal{P} = \left\{\sum_{k=-N}^{N} a_k z^k : N \in \mathbb{N},\ a_k \in \mathbb{C}\right\}$. Apply the complex Stone–Weierstrass theorem to show that $\mathcal{P}$ is uniformly dense in $C(\mathbb{T})$. Conclude that $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$.

This problem is a detailed version of Exercise 2.1, where the two density facts were only quoted.

Solution. **(a)** For $k, j \in \mathbb{Z}$, using the parametrization $z = e^{ix}$,

$$\langle z^k, z^j \rangle = \int_{\mathbb{T}} z^k \bar{z}^j \, dm = \frac{1}{2\pi} \int_{-\pi}^{\pi} e^{i(k-j)x} dx = \begin{cases} 1, & k = j, \\ \dfrac{1}{2\pi}\left[\dfrac{e^{i(k-j)x}}{i(k-j)}\right]_{-\pi}^{\pi} = 0, & k \neq j, \end{cases}$$

since $e^{i(k-j)x}$ is 2π-periodic. Each z^k is bounded, hence in L^2. So $\{z^k\}$ is an ONS.

(b) Write $\ell = x_1 - x_0 \in (0, 2\pi)$ and let $0 < \varepsilon < \frac{2\pi - \ell}{2}$, so that the two "ramps" added below do not overlap on the other side of the circle. Define a function ψ_ε on $[x_0 - \varepsilon,\ x_0 - \varepsilon + 2\pi)$ (one full turn) by

$$\psi_\varepsilon(t) = \begin{cases} \dfrac{t - (x_0 - \varepsilon)}{\varepsilon}, & x_0 - \varepsilon \le t \le x_0, \\ 1, & x_0 \le t \le x_1, \\ \dfrac{(x_1 + \varepsilon) - t}{\varepsilon}, & x_1 \le t \le x_1 + \varepsilon, \\ 0, & x_1 + \varepsilon \le t < x_0 - \varepsilon + 2\pi, \end{cases}$$

and set $\varphi_\varepsilon(e^{it}) = \psi_\varepsilon(t)$. The pieces agree at the break points, and ψ_ε tends to 0 at both ends of the period interval, so φ_ε is a well-defined continuous function on $\mathbb{T}$ with $0 \le \varphi_\varepsilon \le 1$ and $\varphi_\varepsilon = \chi_I$

outside the two arcs $J_\varepsilon^- = \{e^{it} : x_0 - \varepsilon < t < x_0\}$ and $J_\varepsilon^+ = \{e^{it} : x_1 < t < x_1 + \varepsilon\}$. On these arcs $|\varphi_\varepsilon - \chi_I| \le 1$. Hence

$$\|\varphi_\varepsilon - \chi_I\|_{L^2}^2 = \int_{J_\varepsilon^- \cup J_\varepsilon^+} |\varphi_\varepsilon - \chi_I|^2 \, dm \le m(J_\varepsilon^-) + m(J_\varepsilon^+) = \frac{2\varepsilon}{2\pi} = \frac{\varepsilon}{\pi} \xrightarrow{\varepsilon \to 0^+} 0.$$

(The same construction works for open or half-open arcs, since endpoints have measure 0; for the full circle $\chi_{\mathbb{T}} = 1$ is already continuous.)

(c) We use the definition of Lebesgue measure through the outer measure: in the parametrization $t \in [-\pi, \pi)$,

$$m(A) = \inf\Big\{\sum_{j=1}^{\infty} m(J_j) : \ J_j \text{ open arcs}, \ A \subseteq \bigcup_j J_j\Big\} \qquad \text{for measurable } A \subseteq \mathbb{T}.$$

Let $\varepsilon > 0$. Choose open arcs $J_1, J_2, \dots$ covering A with $\sum_j m(J_j) < m(A) + \frac{\varepsilon}{2}$, and put $G = \bigcup_j J_j \supseteq A$. Then

$$m(G \setminus A) = m(G) - m(A) \le \sum_j m(J_j) - m(A) < \tfrac{\varepsilon}{2}.$$

As $\sum_j m(J_j) < \infty$, choose n with $\sum_{j>n} m(J_j) < \frac{\varepsilon}{2}$ and let $U = J_1 \cup \cdots \cup J_n$, a finite union of arcs. Since $A \triangle U = (A \setminus U) \cup (U \setminus A)$ is a disjoint union,

$$m(A \triangle U) = m(A \setminus U) + m(U \setminus A) \le m\Big(\bigcup_{j>n} J_j\Big) + m(G \setminus A) < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon,$$

because $A \setminus U \subseteq G \setminus U \subseteq \bigcup_{j>n} J_j$ and $U \setminus A \subseteq G \setminus A$. Finally, $|\chi_U - \chi_A| = \chi_{A \triangle U}$ (it equals 1 exactly at points lying in one of the sets but not the other), so

$$\|\chi_U - \chi_A\|_{L^2}^2 = \int_{\mathbb{T}} \chi_{A \triangle U} \, dm = m(A \triangle U) < \varepsilon.$$

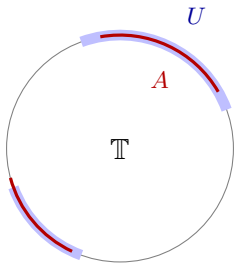


Figure 12: A measurable set A (here two arcs, but in general arbitrary) and a finite union of arcs U with small symmetric difference $A \triangle U$.

(d) Let $\overline{C}$ denote the closure of $C(\mathbb{T})$ in $L^2(\mathbb{T}, m)$; it is a closed linear subspace (closure of a subspace). We show $\overline{C} = L^2$.

Step 1: indicators of arcs. If I is an arc of length $< 2\pi$, then $\chi_I \in \overline{C}$ by (b) (open, closed or half-open arcs differ by at most two points, a null set). For $I = \mathbb{T}$, $\chi_{\mathbb{T}} = 1 \in C(\mathbb{T})$.

Step 2: indicators of finite unions of arcs. A finite union of arcs U can be written as a disjoint union of finitely many arcs $I_1, \dots, I_r$ (up to finitely many points): cut $\mathbb{T}$ at all endpoints and keep the pieces lying in U. Then $\chi_U = \sum_i \chi_{I_i}$ a.e., which lies in $\overline{C}$ by Step 1, as $\overline{C}$ is a linear subspace.

Step 3: indicators of measurable sets. By (c), χ_A is an L^2-limit of functions χ_U from Step 2, so $\chi_A \in \overline{C}$ since $\overline{C}$ is closed.

Step 4: simple functions. A simple function $s = \sum_{j=1}^{r} c_j \chi_{A_j}$ ($c_j \in \mathbb{C}$, A_j measurable) lies in $\overline{C}$ by linearity.

Step 5: all of L^2. Let $f \in L^2$. Writing $f = (u^+ - u^-) + i(v^+ - v^-)$ with $u = \operatorname{Re} f$, $v = \operatorname{Im} f$, each of the four non-negative parts is the pointwise limit of a nondecreasing sequence of simple functions (e.g. $s_n = \min(n, 2^{-n}\lfloor 2^n u^+ \rfloor)$). This gives simple functions $s_n \to f$ pointwise a.e. with $|s_n| \le |f|$ (the standard construction can be arranged so that each part is bounded by the corresponding part of f). Then $|f - s_n|^2 \le (2|f|)^2 = 4|f|^2 \in L^1$ and $|f - s_n|^2 \to 0$ a.e., so by DCT $\|f - s_n\|_{L^2} \to 0$. Hence $f \in \overline{C}$.

Therefore $C(\mathbb{T})$ is dense in $L^2(\mathbb{T}, m)$.

(e) *Stone–Weierstrass* (Theorem 2.2): $\mathbb{T}$ is a compact metric space, and $\mathcal{P} \subset C(\mathbb{T})$ satisfies:

- $\mathcal{P}$ is a subalgebra: it is clearly a linear subspace, and the product of two trigonometric polynomials is one, since $z^k z^j = z^{k+j}$;
- $\mathcal{P}$ contains the constants ($N = 0$);
- $\mathcal{P}$ separates points: $p(z) = z$ belongs to $\mathcal{P}$ and is injective;
- $\mathcal{P}$ is closed under complex conjugation: on $\mathbb{T}$ we have $\overline{z} = z^{-1}$, so $\overline{\sum_{k=-N}^{N} a_k z^k} = \sum_{k=-N}^{N} \overline{a_k} z^{-k} = \sum_{k=-N}^{N} \overline{a_{-k}} z^k \in \mathcal{P}$.

(The last condition is essential: the algebra of polynomials $\sum_{k\ge 0} a_k z^k$ satisfies the first three but is not dense in $C(\mathbb{T})$ – e.g. $\overline{z}$ is not a uniform limit of them, since $\int_{\mathbb{T}} p(z)\, z\, dm = 0$ for such p, while $\int_{\mathbb{T}} \overline{z}\, z\, dm = 1$.) Hence $\mathcal{P}$ is dense in $(C(\mathbb{T}), \|\cdot\|_\infty)$.

ONB. By (a), $\{z^k\}$ is an ONS. Let $f \in L^2(\mathbb{T}, m)$ and $\varepsilon > 0$. By (d) there is $g \in C(\mathbb{T})$ with $\|f - g\|_{L^2} < \varepsilon$, and by the above there is $p \in \mathcal{P}$ with $\|g - p\|_\infty < \varepsilon$. As $m(\mathbb{T}) = 1$, $\|g - p\|_{L^2} = \left(\int |g - p|^2 dm\right)^{1/2} \le \|g - p\|_\infty$, hence

$$\|f - p\|_{L^2} \le \|f - g\|_{L^2} + \|g - p\|_{L^2} < 2\varepsilon.$$

Since $p \in \operatorname{span}\{z^k : k \in \mathbb{Z}\}$, this shows $\operatorname{clos}\operatorname{span}\{z^k\} = L^2(\mathbb{T}, m)$, i.e. $\{z^k : k \in \mathbb{Z}\}$ is an ONB of $L^2(\mathbb{T}, m)$. (Alternatively, use Fejér's theorem instead of Stone–Weierstrass, Corollaries 2.18 and 2.19.)

3.2 List 2 (20. 10. 2026)

The problems of this sheet are numbered as on the sheet itself (continuing from List 1).

Problem 4. Consider

$$H = L^2([-\pi, \pi], dm), \qquad f(x) = x^2,$$

where m is as defined previously.

(a) Compute $\hat{f}(k)$ for every $k \in \mathbb{Z}$.

(b) Compute $\|f\|_{L^2}^2$, and apply Parseval's identity to establish $\displaystyle\sum_{k=1}^{\infty} \frac{1}{k^4} = \frac{\pi^4}{90}$.

(c) Prove that the Fourier series of f converges absolutely and uniformly on $[-\pi, \pi]$, and show that its sum equals f everywhere on this interval. *Hint.* Use Parseval's identity and the continuity of f.

Here $dm = \frac{dx}{2\pi}$ is the normalized measure, i.e. we identify $[-\pi, \pi)$ with $\mathbb{T}$ via $x \mapsto e^{ix}$ as in Section 2; so $\langle g, h\rangle = \frac{1}{2\pi}\int_{-\pi}^{\pi} g\,\overline{h}\,dx$, the ONB is $\{e^{ikx}\}_{k\in\mathbb{Z}}$ (Problem 3) and $\hat{g}(k) = \frac{1}{2\pi}\int_{-\pi}^{\pi} g(x)e^{-ikx}dx$. This is the same setting as in Example 1.31, where $f(x) = x$ gave $\sum 1/k^2$; one more power of x gives $\sum 1/k^4$.

Solution. **(a)** For $k=0$: $\hat f(0)=\frac{1}{2\pi}\int_{-\pi}^{\pi}x^2dx=\frac{1}{2\pi}\cdot\frac{2\pi^3}{3}=\frac{\pi^2}{3}$.

For $k\neq 0$: since x^2 is even and $\sin kx$ is odd, $\int_{-\pi}^{\pi}x^2e^{-ikx}dx=\int_{-\pi}^{\pi}x^2\cos kx\,dx$. Integrate by parts twice, using $\sin k\pi=0$ and $\cos(\pm k\pi)=(-1)^k$:

$$\int_{-\pi}^{\pi}x^2\cos kx\,dx=\Big[x^2\frac{\sin kx}{k}\Big]_{-\pi}^{\pi}-\frac{2}{k}\int_{-\pi}^{\pi}x\sin kx\,dx=-\frac{2}{k}\int_{-\pi}^{\pi}x\sin kx\,dx,$$

$$\int_{-\pi}^{\pi}x\sin kx\,dx=\Big[-x\frac{\cos kx}{k}\Big]_{-\pi}^{\pi}+\frac{1}{k}\int_{-\pi}^{\pi}\cos kx\,dx=-\frac{\pi(-1)^k+\pi(-1)^k}{k}+0=-\frac{2\pi(-1)^k}{k}.$$

Hence $\int_{-\pi}^{\pi}x^2\cos kx\,dx=\frac{4\pi(-1)^k}{k^2}$ and

$$\boxed{\hat f(0)=\frac{\pi^2}{3},\qquad \hat f(k)=\frac{2(-1)^k}{k^2}\quad(k\neq 0).}$$

(Check: $\hat f(-k)=\hat f(k)\in\mathbb{R}$, as it must be for a real even function.)

(b) $\|f\|_{L^2}^2=\frac{1}{2\pi}\int_{-\pi}^{\pi}x^4dx=\frac{1}{2\pi}\cdot\frac{2\pi^5}{5}=\frac{\pi^4}{5}$. By Parseval's identity (3) for the ONB $\{e^{ikx}\}_{k\in\mathbb{Z}}$ (enumerated as a sequence, as in Example 1.31),

$$\frac{\pi^4}{5}=\sum_{k\in\mathbb{Z}}|\hat f(k)|^2=\frac{\pi^4}{9}+\sum_{k\neq 0}\frac{4}{k^4}=\frac{\pi^4}{9}+8\sum_{k=1}^{\infty}\frac{1}{k^4}.$$

Therefore

$$\sum_{k=1}^{\infty}\frac{1}{k^4}=\frac{1}{8}\Big(\frac{\pi^4}{5}-\frac{\pi^4}{9}\Big)=\frac{1}{8}\cdot\frac{4\pi^4}{45}=\frac{\pi^4}{90}.$$

(c) The Fourier series of f is

$$\sum_{k\in\mathbb{Z}}\hat f(k)e^{ikx}=\frac{\pi^2}{3}+\sum_{k\neq 0}\frac{2(-1)^k}{k^2}e^{ikx}=\frac{\pi^2}{3}+4\sum_{k=1}^{\infty}\frac{(-1)^k}{k^2}\cos kx,$$

pairing the terms k and $-k$ ($e^{ikx}+e^{-ikx}=2\cos kx$).

Absolute and uniform convergence. For every x, $|\hat f(k)e^{ikx}|=|\hat f(k)|=M_k$, where $\sum_k M_k=\frac{\pi^2}{3}+4\sum_{k\geq 1}\frac{1}{k^2}=\frac{\pi^2}{3}+\frac{4\pi^2}{6}=\pi^2<\infty$ (Example 1.31). By the Weierstrass M-test (Theorem 3.3) the series converges absolutely and uniformly on $[-\pi,\pi]$ (in fact on $\mathbb{R}$), and its sum

$$g(x)=\sum_{k\in\mathbb{Z}}\hat f(k)e^{ikx}$$

does not depend on the order of summation (in particular $g=\lim_N S_N(f)$ uniformly) and is continuous, as a uniform limit of the continuous functions $S_N(f)$. It is also 2π-periodic, so $g(-\pi)=g(\pi)$.

The sum is f almost everywhere. Since $m([-\pi,\pi])=1$, $\|h\|_{L^2}\leq\sup|h|$, so $S_N(f)\to g$ also in L^2. On the other hand $S_N(f)\to f$ in L^2 by Riesz–Fischer (**). Limits in L^2 are unique, so $f=g$ in L^2, i.e. $\|f-g\|_{L^2}=0$. In the language of the hint: by continuity of the inner product, $\widehat{g}(k)=\lim_N\langle S_N(f),e^{ikx}\rangle=\hat f(k)$ for all k, so Parseval's identity applied to $f-g$ gives $\|f-g\|_{L^2}^2=\sum_k|\hat f(k)-\hat g(k)|^2=0$. Either way, $f=g$ almost everywhere.

Everywhere. Both f and g are continuous on $[-\pi,\pi]$. If $f(x_0)\neq g(x_0)$ for some x_0, then by continuity $|f-g|>\delta>0$ on some interval $J\ni x_0$ of positive length (relative to $[-\pi,\pi]$), and $\|f-g\|_{L^2}^2\geq\delta^2 m(J)>0$, a contradiction. Hence for *every* $x\in[-\pi,\pi]$

$$x^2=\frac{\pi^2}{3}+4\sum_{k=1}^{\infty}\frac{(-1)^k}{k^2}\cos kx.\tag{6}$$

This works at the endpoints too precisely because the periodic extension of x^2 is continuous: $f(-\pi)=f(\pi)$. Compare with $f(x)=x$ in Example 1.31, whose periodic extension jumps at $\pm\pi$, and whose Fourier series there converges to $0\neq\pm\pi$.

Background 3. (Weierstrass M-test)

Let $u_k\colon E\to\mathbb{C}$ be functions on a set E with $\sup_E|u_k|\le M_k$ and $\sum_k M_k<\infty$. Then $\sum_k u_k$ converges absolutely at every point and uniformly on E: the partial sums are uniformly Cauchy, since $\sup_E|\sum_{k=n+1}^{n'}u_k|\le\sum_{k>n}M_k\to 0$. If the u_k are continuous, so is the sum (a uniform limit of continuous functions is continuous). Absolute convergence also implies that every rearrangement has the same sum.

Remark 3.4. Evaluating (6) at special points gives more series for free:

- $x=\pi$: $\pi^2=\frac{\pi^2}{3}+4\sum_k\frac{1}{k^2}$, i.e. $\sum_{k\ge1}\frac{1}{k^2}=\frac{\pi^2}{6}$ again (now from pointwise convergence instead of Parseval);
- $x=0$: $0=\frac{\pi^2}{3}+4\sum_k\frac{(-1)^k}{k^2}$, i.e. $\sum_{k\ge1}\frac{(-1)^{k+1}}{k^2}=\frac{\pi^2}{12}$.

The general principle behind (c): *if $\sum_k|\hat f(k)|<\infty$ and f is continuous, then the Fourier series converges to f uniformly.* Smoothness of f (here: f is continuous and piecewise C^1 as a function on $\mathbb{T}$) gives decay of $\hat f(k)$ (here $\sim k^{-2}$), and decay gives good convergence. For $f(x)=x$ the coefficients only decay like k^{-1}, which is not summable, and uniform convergence fails.

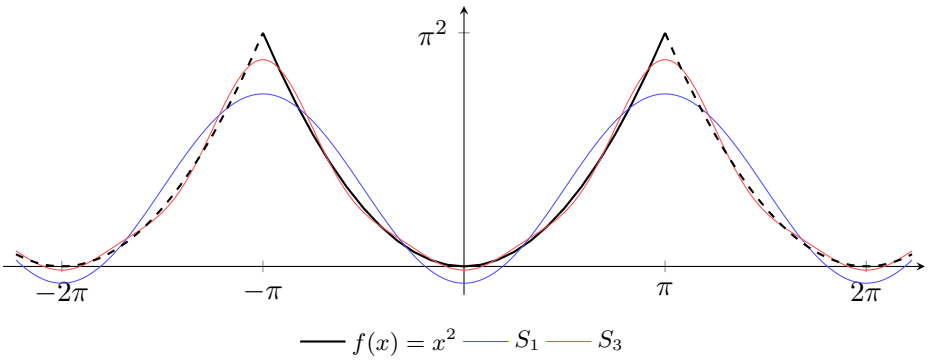


Figure 13: $f(x)=x^2$ on $[-\pi,\pi]$ (solid), its 2π-periodic extension (dashed) and the partial sums $S_N=\frac{\pi^2}{3}+4\sum_{k=1}^N\frac{(-1)^k}{k^2}\cos kx$. The extension is continuous, and already S_3 is uniformly close to f, in contrast with $f(x)=x$ (Example 1.31).

Problem 5. Let $f,g\in L^2(\mathbb{T},m)$. Prove that, for every $n\in\mathbb{Z}$,

$$\widehat{fg}(n)=\sum_{k\in\mathbb{Z}}\hat f(k)\hat g(n-k),$$

and that the series converges absolutely.

Solution. Step 0: everything is well defined. By Cauchy–Schwarz, $\int_{\mathbb{T}}|fg|\,dm\le\|f\|_{L^2}\|g\|_{L^2}<\infty$, so $fg\in L^1(\mathbb{T},m)$ and $\widehat{fg}(n)=\int_{\mathbb{T}}fg\,\overline{z}^n\,dm$ is defined (although fg need not be in L^2; cf. Problem 2(a)).

Step 1: absolute convergence. By Parseval, $\sum_k|\hat f(k)|^2=\|f\|^2<\infty$ and, substituting $j=n-k$ (a bijection of $\mathbb{Z}$), $\sum_k|\hat g(n-k)|^2=\sum_j|\hat g(j)|^2=\|g\|^2$. Cauchy–Schwarz in $\ell^2(\mathbb{Z})$ gives, for every finite $F\subset\mathbb{Z}$,

$$\sum_{k\in F}|\hat f(k)||\hat g(n-k)|\le\Big(\sum_{k\in F}|\hat f(k)|^2\Big)^{1/2}\Big(\sum_{k\in F}|\hat g(n-k)|^2\Big)^{1/2}\le\|f\|_{L^2}\|g\|_{L^2}.$$

So $\sum_{k\in\mathbb{Z}}|\hat f(k)\hat g(n-k)|\le\|f\|\,\|g\|<\infty$: the series converges absolutely, hence its sum does not depend on the order of summation, and it equals $\lim_{N\to\infty}\sum_{|k|\le N}\hat f(k)\hat g(n-k)$.

Step 2: write $\widehat{fg}(n)$ as an inner product. Put $h(z) = \overline{g(z)}z^n$. Since $|z| = 1$, $|h| = |g|$, so $h \in L^2(\mathbb{T}, m)$, and

$$\widehat{fg}(n) = \int_{\mathbb{T}} f\, g\, \overline{z}^n \, dm = \int_{\mathbb{T}} f\, \overline{\overline{g}\, z^n}\, dm = \langle f, h\rangle.$$

The Fourier coefficients of h are, using $\overline{z} = z^{-1}$ on $\mathbb{T}$,

$$\hat{h}(k) = \int_{\mathbb{T}} \overline{g}\, z^n \overline{z}^k \, dm = \overline{\int_{\mathbb{T}} g\, \overline{z}^n z^k \, dm} = \overline{\int_{\mathbb{T}} g\, \overline{z}^{n-k}\, dm} = \overline{\hat{g}(n-k)}.$$

Step 3: Parseval for inner products. For $u, v \in L^2(\mathbb{T}, m)$,

$$\langle u, v\rangle = \sum_{k\in\mathbb{Z}} \hat{u}(k)\overline{\hat{v}(k)}. \tag{7}$$

Indeed, by Riesz–Fischer $S_N(u) \to u$ and $S_N(v) \to v$ in L^2, so by continuity of the inner product (Lemma 1.28) and orthonormality of $\{z^k\}$,

$$\langle u, v\rangle = \lim_{N\to\infty} \langle S_N(u), S_N(v)\rangle = \lim_{N\to\infty} \sum_{|k|,|j|\le N} \hat{u}(k)\overline{\hat{v}(j)}\langle z^k, z^j\rangle = \lim_{N\to\infty} \sum_{|k|\le N} \hat{u}(k)\overline{\hat{v}(k)}.$$

(Alternatively, (7) follows from (3) by polarization, $\langle u, v\rangle = \frac{1}{4}\sum_{r=0}^{3} i^r \, \|u + i^r v\|^2$.)

Step 4: conclusion. Applying (7) with $u = f$, $v = h$:

$$\widehat{fg}(n) = \langle f, h\rangle = \sum_{k\in\mathbb{Z}} \hat{f}(k)\,\overline{\hat{h}(k)} = \sum_{k\in\mathbb{Z}} \hat{f}(k)\hat{g}(n-k),$$

the series being absolutely convergent by Step 1.

Remark 3.5. The right-hand side is the *convolution* of the sequences $\hat{f}, \hat{g} \in \ell^2(\mathbb{Z})$: $(\hat{f} * \hat{g})(n) = \sum_k \hat{f}(k)\hat{g}(n-k)$. So the problem says $\widehat{fg} = \hat{f} * \hat{g}$: *the Fourier transform turns products into convolutions.* The dual statement, $\widehat{f * g} = \hat{f}\,\hat{g}$ for the convolution of functions on $\mathbb{T}$, is the one that appeared with the Fejér kernel ($\sigma_N(f) = f * F_N$). For trigonometric polynomials the identity is just the rule for multiplying polynomials: the coefficient of z^n in $\big(\sum_k a_k z^k\big)\big(\sum_j b_j z^j\big)$ is $\sum_{k+j=n} a_k b_j$. Step 1 also shows $|\widehat{fg}(n)| \le \|f\|_{L^2}\,\|g\|_{L^2}$, consistent with $|\widehat{fg}(n)| \le \|fg\|_{L^1}$.

Problem 6. Let $BV([a,b])$ denote the set of $\mathbb{C}$-valued functions of bounded variation on $[a,b]$, where

$$\mathrm{Var}_{[a,b]}(f) := \sup_{a=t_0<\cdots<t_N=b} \sum_{j=0}^{N-1} |f(t_{j+1}) - f(t_j)|.$$

(a) Verify that $BV([a,b])$ is a vector space.

(b) Show that every $f \in BV([a,b])$ is bounded. *Hint.* Fix $x \in [a,b]$ and estimate $|f(x) - f(a)|$.

(c) Prove that if $f, g \in BV([a,b])$, then $fg \in BV([a,b])$. *Hint.* Use

$$f(t_{j+1})g(t_{j+1}) - f(t_j)g(t_j) = f(t_{j+1})(g(t_{j+1}) - g(t_j)) + g(t_j)(f(t_{j+1}) - f(t_j)).$$

(d) Let $f \in BV([a,b])$ be real-valued. Define

$$V(x) := \mathrm{Var}_{[a,x]}(f), \qquad x \in [a,b].$$

Show that both V and $V - f$ are nondecreasing on $[a,b]$. Deduce the Jordan decomposition theorem mentioned in the lectures.

In the lectures the same quantity was written $V_a^b(f)$; we write $\mathrm{Var}_{[a,b]}(f)$ as on the sheet. For a partition $P = \{a = t_0 < \cdots < t_N = b\}$ write $S(f,P) = \sum_{j=0}^{N-1} |f(t_{j+1}) - f(t_j)|$, so $\mathrm{Var}_{[a,b]}(f) = \sup_P S(f,P)$ and $f \in BV([a,b])$ means $\mathrm{Var}_{[a,b]}(f) < \infty$. We set $\mathrm{Var}_{[a,a]}(f) = 0$.

Solution. **(a)** $BV([a,b])$ is a subset of the vector space of all functions $[a,b]\to\mathbb{C}$; we check that it is a linear subspace. The zero function has $S(0,P)=0$ for all P, so $0\in BV$. Let $f,g\in BV$, $\alpha,\beta\in\mathbb{C}$, and P any partition. By the triangle inequality,

$$\begin{aligned}S(\alpha f+\beta g,P)&=\sum_j|\alpha(f(t_{j+1})-f(t_j))+\beta(g(t_{j+1})-g(t_j))|\\&\le|\alpha|S(f,P)+|\beta|S(g,P)\le|\alpha|\operatorname{Var}(f)+|\beta|\operatorname{Var}(g).\end{aligned}$$

Taking the supremum over P: $\operatorname{Var}(\alpha f+\beta g)\le|\alpha|\operatorname{Var}(f)+|\beta|\operatorname{Var}(g)<\infty$, so $\alpha f+\beta g\in BV$. (In particular Var is a seminorm on BV; it vanishes exactly on constants, and $\|f\|_{BV}=|f(a)|+\operatorname{Var}_{[a,b]}(f)$ is a norm.)

(b) Let $x\in(a,b)$. Then $a<x<b$ is a partition, so

$$|f(x)-f(a)|\le|f(x)-f(a)|+|f(b)-f(x)|=S(f,\{a,x,b\})\le\operatorname{Var}_{[a,b]}(f).$$

For $x=b$ use the partition $a<b$, and for $x=a$ the estimate is trivial. Hence for all $x\in[a,b]$

$$|f(x)|\le|f(a)|+|f(x)-f(a)|\le|f(a)|+\operatorname{Var}_{[a,b]}(f)<\infty,$$

so $\|f\|_\infty:=\sup_{[a,b]}|f|\le|f(a)|+\operatorname{Var}_{[a,b]}(f)$.

(c) Let P be a partition. By the identity in the hint and the triangle inequality,

$$\begin{aligned}S(fg,P)&\le\sum_j|f(t_{j+1})|\,|g(t_{j+1})-g(t_j)|+\sum_j|g(t_j)|\,|f(t_{j+1})-f(t_j)|\\&\le\|f\|_\infty S(g,P)+\|g\|_\infty S(f,P).\end{aligned}$$

Taking the supremum over P and using (b) (which guarantees $\|f\|_\infty,\|g\|_\infty<\infty$),

$$\operatorname{Var}(fg)\le\|f\|_\infty\operatorname{Var}(g)+\|g\|_\infty\operatorname{Var}(f)<\infty.$$

So $BV([a,b])$ is not only a vector space but an *algebra*. (The boundedness in (b) is essential here: this is the only place where it is used.)

(d) *Key inequality.* For $a\le x<y\le b$:

$$V(y)\ge V(x)+|f(y)-f(x)|.\tag{8}$$

Proof: if $x=a$, then $V(a)=0$ and $\{a<y\}$ is a partition of $[a,y]$, so $V(y)\ge|f(y)-f(a)|$. If $x>a$, let $P=\{a=t_0<\cdots<t_N=x\}$ be any partition of $[a,x]$. Then $P'=P\cup\{y\}$ is a partition of $[a,y]$ and

$$S(f,P)+|f(y)-f(x)|=S(f,P')\le V(y).$$

Taking the supremum over P gives (8). In particular (with $y=b$) $V(x)\le V(b)=\operatorname{Var}_{[a,b]}(f)<\infty$, so V is a finite, real-valued function.

V is nondecreasing: by (8), $V(y)-V(x)\ge|f(y)-f(x)|\ge0$ for $x<y$.

$V-f$ is nondecreasing: since f is real, $|f(y)-f(x)|\ge f(y)-f(x)$, so by (8)

$$\big(V(y)-f(y)\big)-\big(V(x)-f(x)\big)=\big(V(y)-V(x)\big)-\big(f(y)-f(x)\big)\ge|f(y)-f(x)|-\big(f(y)-f(x)\big)\ge0.$$

(Similarly $V+f$ is nondecreasing.)

Jordan decomposition (Theorem 2.30): $g\in BV([a,b])$ if and only if $g=g_1-g_2+i(g_3-g_4)$ with $g_1,\dots,g_4\colon[a,b]\to\mathbb{R}$ bounded and nondecreasing.

"⇒": Let $g\in BV([a,b])$ and $u=\operatorname{Re}g$, $v=\operatorname{Im}g$. Since $|\operatorname{Re}w|\le|w|$, $S(u,P)\le S(g,P)$ for every P, so $u\in BV$, and likewise $v\in BV$. By the above, with $V_u(x)=\operatorname{Var}_{[a,x]}(u)$,

$$u=V_u-(V_u-u),$$

a difference of two nondecreasing functions; they are bounded, since a nondecreasing h on $[a,b]$ satisfies $h(a) \le h \le h(b)$. Do the same for v and set $g_1 = V_u$, $g_2 = V_u - u$, $g_3 = V_v$, $g_4 = V_v - v$.

"$\Leftarrow$": A nondecreasing h is in BV: for every partition the sum telescopes, $S(h,P) = \sum_j (h(t_{j+1}) - h(t_j)) = h(b) - h(a)$, so $\mathrm{Var}(h) = h(b) - h(a) < \infty$. By (a), the linear combination $g_1 - g_2 + ig_3 - ig_4$ is in BV.

Remark 3.6. (a) The decomposition is far from unique (add the same nondecreasing function to g_1 and g_2), but V is in a sense optimal: one can show that $\mathrm{Var}_{[x,y]}(f) = V(y) - V(x)$ (additivity of the variation), so for real f the two nondecreasing functions $\frac{V+f}{2}$ and $\frac{V-f}{2}$ are the "positive and negative variation" of f.

(b) Using (8) instead of additivity makes the proof shorter than the one in the lectures, which used $f(y) - f(x) \le V_x^y(f) = V(y) - V(x)$.

(c) Consequences: by the Jordan decomposition and the fact that monotone functions have at most countably many jumps and are differentiable a.e. (Lebesgue), the same holds for BV functions. This is the input for Theorem 2.32 and Hardy's theorem (Theorem 2.36).

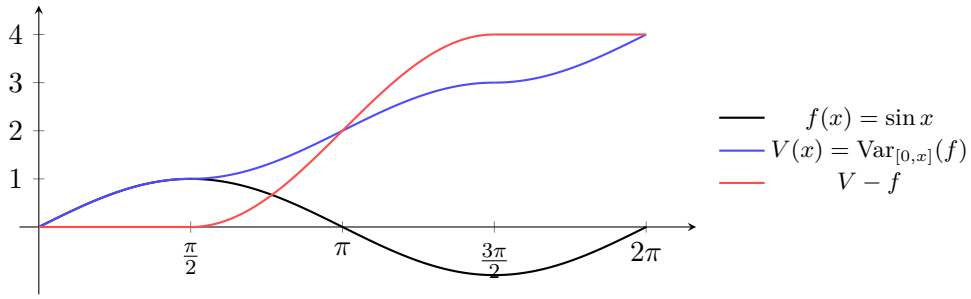


Figure 14: Jordan decomposition of $f(x) = \sin x$ on $[0, 2\pi]$: V grows wherever f moves (up or down), $V - f$ grows only where f decreases, and $f = V - (V - f)$. Here $\mathrm{Var}_{[0,2\pi]}(\sin) = 4$.